

CATHOLIC UNIVERSITY OF EASTERN AFRICA



**STUDENTS' GRADUATION CLEARANCE SYSTEM FOR TANGAZA
UNIVERSITY**

BY

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DECLARATION

Student's Declaration

This research project is my original work and has not been presented for a degree/diploma award in any other institution.

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DEDICATION

This project is dedicated to God, whose endless support, patience, and belief in me made this journey possible, even on the nights it felt like graduation would never come. To my friends and classmates who debugged code with me at 2 a.m., sent motivational memes when deadlines loomed, and reminded me to eat actual food during crunch weeks, this one's for you too.

To my parents, who've shown constant support .

And finally, to my future self: you did it. Now go walk across that stage.

ABSTRACT

Graduation clearance is an important administrative process that ensures final-year students have fulfilled academic and other institutional requirements before being approved for graduation. At Tangaza University, the clearance process has traditionally relied on manual procedures requiring students to visit multiple departments to obtain approvals. This approach is time-consuming, prone to delays, and makes it difficult for students and administrators to track clearance progress effectively.

This project involved the design and development of a web-based Students' Graduation Clearance System (SGCS) for Tangaza University to automate and centralize the graduation clearance process. The system enables students to submit clearance requests, upload supporting documents, monitor clearance progress, and receive notifications. Administrative officers can review and process clearance requests, while authorized management users can monitor clearance activities and generate reports. The system was developed using web technologies and a relational database to provide centralized and structured management of clearance information.

The system was evaluated through functional testing, User Acceptance Testing (UAT), and usability evaluation. UAT involved four final-year students and two administrative staff members, with participants reporting positive experiences with the system. The evaluation produced an overall satisfaction score of 4.3 out of 5, while the clarity of the clearance status tracker received an average score of 4.6 out of 5. The results indicate that the SGCS can improve efficiency, transparency, communication, and tracking of the graduation clearance process compared with the existing manual approach. The system therefore provides a practical digital solution for streamlining graduation clearance at Tangaza University.

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DEFINITION OF KEY TERMS

CSS	Cascading Style Sheets. A style sheet language used to describe the presentation of HTML documents.
DFD	Data Flow Diagram. A graphical representation of how data flows through a system.
ERD	Entity Relationship Diagram. A modelling tool used in database design.
HTML	HyperText Markup Language. The standard language for creating web pages.
ICT	Information and Communication Technology.
JKUAT	Jomo Kenyatta University of Agriculture and Technology.
JS	JavaScript. A scripting language for dynamic web content.
NF	Normal Form. A stage in database normalisation.
PHP	Hypertext Preprocessor. A widely used server-side scripting language for web development.
SGCS	Students' Graduation Clearance System.
SSAD	Structured System Analysis and Design. A system development methodology.
UAT	User Acceptance Testing. The final phase of system testing involving real users.
UI/UX	User Interface / User Experience.
UoN	University of Nairobi.

CHAPTER ONE: INTRODUCTION

1.1 Motivation & Background

This project did not come from a textbook. It came from watching final-year students at Tangaza University run around campus for entire afternoons, trying to get signatures for their graduation clearance, only to be told to come back the next day because an office was closed or a form couldn't be found. Knowing that I would have to go through the exact same process myself is what made this topic personal to me, and not just something I picked for a research paper.

Universities across Kenya are being pushed to modernize the way they run administrative work, as part of the country's wider push toward digital services. Graduation clearance is one of those processes that seems almost designed to benefit from this kind of change, it's repetitive, it's paper-heavy, and it involves the same steps for every single student. Yet at Tangaza University, and many others like it, clearance is still one of the least digitized processes on campus. This research comes from the belief that fixing this doesn't require overhauling the entire university system, a well-built, focused solution can solve a problem that affects every student who wants to graduate.

1.2 Background of the Research

The rapid growth of Information and Communication Technology (ICT) has transformed the way organizations deliver services by improving efficiency, accuracy, and accessibility. In higher education, universities have increasingly adopted digital systems to manage administrative activities such as student registration, fee payment, examination processing, and academic records. Studies by Mwangi and Otieno (2021) and Munyua (2020) indicate that digital systems reduce processing time, improve record accuracy, and enhance service delivery.

Graduation clearance is an important administrative process through which final-year students confirm that they have fulfilled all academic, financial, library, and other institutional obligations before they are approved for graduation. In many universities, including Tangaza University, this process is still conducted manually using paper forms that require students to obtain approvals from multiple departments. This often leads to long

queues, repeated visits, delays, misplaced documents, and difficulty in tracking the clearance progress.

Tangaza University currently lacks an automated graduation clearance system. Students are required to move from one office to another to obtain signatures, making the process time-consuming, inconvenient, and prone to errors. Administrative staff also rely on manual records, making it difficult to monitor student progress and generate reports efficiently.

This study proposes the development of a Students' Graduation Clearance System (SGCS) to automate and centralize the graduation clearance process. The system enables students to submit clearance requests online, track their clearance status in real time, receive in-app notifications, and upload required supporting documents. Departmental officers review and process clearance requests electronically, while the Academic Registrar uploads transcripts and monitors the overall clearance process through centralized reports. The proposed system is expected to improve efficiency, reduce paperwork, minimize delays, and enhance service delivery at Tangaza University.

1.3 Problem Statement

The graduation clearance process at Tangaza University is currently manual and paper-based, requiring final-year students to visit six departments to obtain approval. This often results in delays due to repeated visits, unavailable officers, and the absence of a system for tracking clearance progress. Students are also unable to receive timely updates on outstanding issues or their clearance status.

The manual process also creates administrative challenges, including poor record management, lack of centralized reporting, weak communication between departments, and the risk of lost or inconsistent records. These limitations reduce efficiency, transparency, and accountability, highlighting the need for an automated Students' Graduation Clearance System to streamline and improve the clearance process.

1.4 Aim of the Research

To design and develop a web-based Students' Graduation Clearance System that automates and streamlines the multi-departmental graduation clearance process at Tangaza University.

1.5 Objectives of the Research

The following specific objectives will guide the achievement of the main objective:

1. To review existing literature and related graduation clearance systems to identify best practices, technologies, and research gaps relevant to the proposed system.
2. To analyse the current manual graduation clearance process at Tangaza University through document analysis and questionnaires, identifying its limitations and the requirements for the proposed system.
3. To design the proposed Students' Graduation Clearance System, including the database structure, process flows, and user interface prototypes, guided by the SSAD methodology.
4. To develop the core system functionalities, including the multi-departmental sequential clearance workflow and departmental reporting for the Director and the Academic Registrar; financial clearance approval requires a verified fee-data integration.
5. To implement real-time status tracking and automated notification features that keep students and officers informed at each stage of the clearance process.
6. To test the developed system through developer testing, end-user testing, and heuristic evaluation, and to deploy the system within the Tangaza University environment.

1.6 Justification of the Research

The development of the Students' Graduation Clearance System (SGCS) is justified by the need to improve the efficiency and effectiveness of the graduation clearance process at Tangaza University. According to Mwangi and Otieno (2021), digital administrative systems reduce processing time and improve service delivery. The proposed system will eliminate repetitive manual procedures, reduce paperwork, and enable students to complete clearance online.

The SGCS will also reduce the delays and stress associated with the current manual process by providing real-time clearance tracking and automated notifications, as highlighted by Kariuki (2020). In addition, the system will improve transparency, accountability, and record management by maintaining centralized digital records and generating accurate reports for university management.

Furthermore, the proposed system supports Kenya's digital transformation agenda and provides a sustainable, cost-effective solution that can be easily maintained and expanded to meet future institutional needs.

1.7 Scope of the Research

This study focuses on the development of a Students' Graduation Clearance System (SGCS) for Tangaza University, Nairobi. The system will automate the graduation clearance process across six departments: the Director, Library, Dean's Office, University Store, Finance Office, and Academic Registrar.

The system is used by final-year undergraduate students and departmental officers. Its key features include online clearance initiation, sequential departmental approvals, supporting-document checks, Academic Registrar transcript upload, real-time status tracking, in-app notifications, report generation, and issuance of a digital final approval.

The study is limited to the graduation clearance process and does not cover other university functions such as student registration, examination management, or fee payment processing.

1.8 Research Organisation

This research is organized into chapters. Chapter Two presents a review of five related systems from global and local contexts, a historical overview of student clearance systems in higher education, emerging trends in the field, and the research gap this project addresses. Chapter Three describes the methodology that will guide the development of the system, covering requirement elicitation, system analysis, design, implementation, testing, and deployment, with all relevant analytical diagrams and design artefacts included.

CHAPTER TWO: REVIEW OF RELATED WORK

2.1 Chapter Introduction

This chapter reviews literature and related systems relevant to the proposed Students' Graduation Clearance System (SGCS). It examines the evolution of university administrative systems, reviews related graduation clearance systems, compares their strengths and weaknesses, and discusses emerging trends. The chapter concludes with a summary of the findings that inform the design of the proposed system.

2.2 History of the Research Topic

The automation of university administrative processes began in the 1980s with computer-based student information systems used mainly for record management by administrators. With the growth of the internet in the 1990s, universities introduced online student portals, allowing students to access services such as registration and academic records.

In the 2000s, many universities adopted integrated management systems that combined registration, finance, library, and academic services. However, adoption in sub-Saharan Africa was slower due to limited ICT infrastructure, high costs, and inadequate technical expertise (Wanyembi, 2018).

In Kenya, the adoption of digital university systems accelerated in the early 2010s, with institutions such as the University of Nairobi, Kenyatta University, and JKUAT implementing online student portals. Despite these developments, graduation clearance has remained largely manual in many institutions, including Tangaza University, highlighting the need for an automated graduation clearance system.

2.3 Review of Related Prototypes/Systems (Global to Local)

2.3.1 Case 1: University of Nairobi (UoN) Student Clearance Portal

The University of Nairobi integrated an online clearance portal into its Student Information System, allowing students to submit clearance requests electronically while departments process approvals online. The system also generates a clearance certificate after all approvals are completed.

Its main strength is integration with the university's student database, which reduces duplicate data entry and allows the Academic Registrar to monitor clearance progress centrally. However, Omondi (2020) noted that the system is not mobile-responsive and only notifies students after final clearance, limiting real-time tracking. The proposed SGCS addresses these limitations by providing a mobile-friendly interface and real-time departmental status updates.

Table 1: Comparison of Proposed System with Case 1 (University of Nairobi)

S/N	Proposed System (SGCS)	Case 1: UoN Clearance Portal
1	Mobile-responsive design accessible via smartphone	Not mobile-responsive
2	Notification at each departmental approval	Email notification only on final clearance
3	Real-time per-department status visible to student	Status not visible between submissions
4	Sequential approval workflow with six department	Limited departmental workflow configuration

2.3.2 Case 2: Catholic University of Eastern Africa (CUEA) Student Portal

The Catholic University of Eastern Africa (CUEA) provides an online graduation clearance module within its student portal, allowing students to initiate and track their clearance process. Departmental officers update clearance records, and students can view their clearance status through a centralized dashboard.

A key strength of the system is its digital clearance workflow, which improves visibility and reduces reliance on paper-based processes. However, the system does not provide automated notifications for departmental approvals. The proposed SGCS addresses these limitations by providing real-time notifications and integrated transcript management.

Table 2: Comparison of Proposed System with Case 2 (Catholic University of Eastern Africa)

S/N	Proposed System (SGCS)	Case 2: CUEA Student Portal
1	Automated notification to student at each departmental approval stage	No automated per-stage notifications; student must check status manually
2	Transcript request and upload module integrated within the clearance workflow	Transcript handling not integrated within the clearance module
3	Sequential workflow with defined approval order enforced by the system	Departmental clearance updates managed independently without enforced sequence
4	Designed specifically for the Tangaza University clearance structure and roles	Generalised institutional portal not tailored to a specific clearance workflow

2.3.3 Case 3: Strathmore University Clearance System

Strathmore University provides an online graduation clearance module through its student portal, enabling students to submit clearance requests electronically and receive departmental feedback. The system also maintains a digital archive of clearance records, improving record management (Strathmore ICT Report, 2021).

Its strengths include an intuitive user interface and efficient digital record management. However, it does not send reminder notifications for pending clearance tasks, and its proprietary design limits adaptability to other institutions. The proposed SGCS addresses these limitations by providing automated reminders and using widely adopted technologies that are easier to maintain and customize.

Table 3: Comparison of Proposed System with Case 3 (Strathmore University)

S/N	Proposed System (SGCS)	Case 3: Strathmore System
1	Automated reminder notifications for pending clearance items	No automated reminders for students with pending items
2	Built on standard, widely documented technologies	Proprietary, closed system
3	Escalation notification if a department delays approval	No escalation mechanism for delayed approvals
4	Digital archive of all clearance records	Digital archiving available
5	Adaptable to other institutions with minimal modification	Not adaptable to other institutions

2.3.4 Case 4: Moi University Student Clearance System

Moi University integrated a graduation clearance module into its Student Information System, with direct integration to the fee management system. This automatically prevents students with outstanding fee balances from completing the clearance process.

Its main strength is automated financial clearance verification, which improves accuracy and reduces manual work. However, Ruto and Sang (2022) reported poor performance during peak clearance periods and the absence of a mobile-friendly interface. The proposed SGCS addresses these limitations by providing responsive design and efficient financial clearance processing.

Table 4: Comparison of Proposed System with Case 4 (Moi University)

S/N	Proposed System (SGCS)	Case 4: Moi University SIS
1	Mobile-responsive interface accessible on smartphones	No mobile-responsive interface
2	Designed for reliable performance under concurrent student load	Slowdowns reported during peak graduation season

S/N	Proposed System (SGCS)	Case 4: Moi University SIS
3	Finance Officer verification step to block clearance for fee arrears	Automated fee blocking integrated with fee system
4	Reporting dashboard for Director and Registrar	Limited reporting features
5	Standard server-side scripting without complex infrastructure	Complex server infrastructure required

2.3.5 Case 5: JKUAT Online Clearance Module

JKUAT provides an online graduation clearance module that allows students to upload supporting documents and enables officers to review and process clearance requests electronically. This has reduced the need for physical visits to administrative offices (Ngugi, 2021).

Its strengths include online document submission and electronic approval processing. However, the system lacks automatic reminders for delayed approvals and still requires students to collect the final clearance document physically. The proposed SGCS addresses these limitations by providing automated notifications and generating a downloadable digital graduation clearance letter.

Table 5: Comparison of Proposed System with Case 5 (JKUAT)

S/N	Proposed System (SGCS)	Case 5: JKUAT Module
1	Automated escalation notification if a department delays beyond a defined period	No escalation mechanism for delayed departmental approvals
2	Supporting document upload for officer verification	Document upload available

S/N	Proposed System (SGCS)	Case 5: JKUAT Module
3	Real-time approval status visible to student per department	Status display is limited
4	Full audit trail maintained in the centralised database	Basic audit trail only

2.4 Emerging Trends and Patterns in the Research Area

2.4.1 Mobile-First Design

With the increasing use of smartphones among students, universities are adopting mobile-friendly systems. While the University of Nairobi and Moi University lack fully responsive interfaces, JKUAT supports access across different devices. The proposed SGCS will use a responsive design to ensure accessibility on smartphones, tablets, and computers.

2.4.2 Cloud-Based and Centralized Systems

Universities are increasingly adopting centrally hosted web systems that allow users to access services remotely. CUEA and KUCCPS demonstrate the benefits of cloud-based platforms by providing reliable access to services from any internet-connected device. Similarly, the proposed SGCS will be a web-based system accessible both on and off campus.

2.4.3 Automation of Administrative Workflows

Modern university systems automate administrative processes such as approvals, notifications, and document handling. Although Strathmore University and JKUAT provide online clearance processing, they lack features such as automated reminders and escalation notifications. The proposed SGCS will automate the clearance workflow through sequential approvals, real-time notifications, and reminders for pending requests.

2.4.4 Artificial Intelligence (AI)

Universities are beginning to explore AI technologies such as chatbots to improve student support and automate routine inquiries. Although none of the reviewed Kenyan university systems currently use AI for graduation clearance, this trend indicates future opportunities. While AI will not be included in the initial SGCS, the system will be designed to support future enhancements.

2.5 Research Gap

The reviewed systems show a gap for a focused, web-based graduation-clearance system for Tangaza University that combines sequential departmental processing, in-app status notifications, required supporting-document checks, role-based access, and final approval in one workflow.

2.6 Chapter Summary

This chapter reviewed five related graduation clearance systems and examined their strengths, weaknesses, and relevance to the proposed SGCS. It also discussed the evolution of university administrative systems and identified key trends, including mobile-first design, cloud-based systems, automated workflows, and AI-assisted services. The review revealed common limitations such as poor mobile support, limited notifications, and reliance on manual processes. These findings informed the design of the proposed Students' Graduation Clearance System, which addresses these gaps. The next chapter presents the methodology used to develop the system.

CHAPTER THREE: METHODOLOGY

3.1 Chapter Introduction

This chapter describes the methodology that will be used to research, design, develop, test, and deploy the proposed Students' Graduation Clearance System (SGCS). It explains the methods and techniques to be applied in each phase of the project, from requirements gathering to system implementation and deployment. The chapter also justifies the selected approaches and technologies to ensure they are suitable for the project objectives and the Tangaza University environment.

3.2 Methodology for Literature Review

The literature review for this study will be conducted through a structured and systematic search of academic databases and institutional repositories. The materials to be reviewed will include peer-reviewed journal articles, conference proceedings, theses, institutional ICT reports, and credible online publications. Materials published between 2016 and 2024 will be prioritised to ensure relevance, although foundational works published prior to this range may be included where they provide essential historical context. The primary databases to be consulted include Google Scholar, ResearchGate, African Journals Online (AJOL), and the institutional repositories of Kenyan universities. The search keywords to be used include: 'student clearance system,' 'graduation clearance automation,' 'university management information system,' 'Django web application,' 'academic portal Kenya,' and 'student information system Africa.' Inclusion criteria will require that materials are relevant to the research topic, published by credible sources, and accessible in full text. Materials that do not meet these criteria will be excluded.

3.3 Methodology for Requirement Specification, Data Collection & Analysis

Requirement elicitation is the process of identifying, in precise and practical terms, what the proposed system must do, as determined by the people who will use it. For the SGCS, two complementary data collection methods will be used: document analysis of official Tangaza University administrative forms and questionnaires administered to final-year students. Each method will be suited to a different type of information and a different source of data.

3.3.1 Document Analysis

Document analysis is a research method that involves examining existing official documents to obtain information relevant to a study.

In this study, two official Tangaza University documents will be analyzed. The first is the Application for Graduation Form, which contains student information such as registration number, programme, institute, contact details, graduation details, and the Academic Programme Leader's approval. Analysis of this form will identify the student data required in the proposed SGCS and confirm that academic eligibility is verified before the clearance process begins.

The second document is the Student Clearance Form, which is currently used to process graduation clearance manually. It requires approvals from six departments: the Director, Library, Dean of Students, University Store, Finance Office, and Academic Registrar. Each department records the student's clearance status, comments, and authorizing signature. Analysis of this form will define the sequential clearance workflow, departmental approval process, and the data required at each stage.

Document analysis is appropriate because these official documents provide an accurate representation of the current graduation clearance process. The findings will guide the design of the database, workflow, and data requirements of the proposed Students' Graduation Clearance System.

3.3.2 Questionnaires

A questionnaire is a structured data collection tool that uses standardized questions to gather information from respondents. It enables consistent collection and analysis of data from a large group of participants.

In this study, questionnaires will be administered to 20 final-year students selected through simple random sampling. The questionnaire will contain both closed-ended and open-ended questions focusing on students' experiences with the current graduation clearance process, the challenges they face, and the features they would like in an online clearance system. It will be distributed using Google Forms, and the responses will be analyzed using frequency tables, percentages, and charts where appropriate.

Questionnaires are appropriate for this study because they allow data to be collected efficiently from many respondents within a short time while ensuring that the proposed Students' Graduation Clearance System reflects the needs and opinions of its intended users.

3.4 Methodology for System Analysis (Current System)

System analysis is the process of studying the existing system to understand how it operates and to identify the requirements for an improved system. In this study, the SSAD methodology will be used to analyze the current manual graduation clearance process at Tangaza University.

3.4.1 Context Diagram

A Context Diagram is a high-level diagram that shows the system, its external entities, and the flow of information between them. It will capture the interactions between the current graduation clearance process and external users such as students, departmental officers, and the Academic Registrar. It is chosen because it clearly defines the system boundaries and the main sources and destinations of data.

3.4.2 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) illustrates how data moves through a system, the processes involved, data stores, and external entities. It will capture the flow of information in the current manual clearance process, including how clearance requests are processed across departments. It is chosen because it provides a clear understanding of the workflow and helps identify inefficiencies in the existing system.

3.4.3 System Flowchart

A System Flowchart is a graphical representation of the sequence of activities performed within a system. It will illustrate the step-by-step graduation clearance process from application to final approval, including decision points and departmental approvals. It is chosen because it clearly shows the operational flow of the current system, making it easier to identify areas requiring improvement.

3.5 Methodology for System Design (Proposed System)

System design is the process of transforming the system requirements identified during analysis into a detailed blueprint for developing the proposed Students' Graduation Clearance System (SGCS). It defines how the system will be structured, how users will interact with it, and how data will be processed and stored.

3.5.1 Database Design

Database design is the process of organizing data into related tables to ensure efficient storage and retrieval. An Entity Relationship Diagram (ERD) will be used to define entities such as students, clearance requests, departments, officers, and their relationships. The database will be normalized to Third Normal Form (3NF) to eliminate redundancy and maintain data integrity. This approach is chosen because it ensures accurate, consistent, and efficient data management.

3.5.2 Context Diagram

A Context Diagram provides a high-level view of the proposed system by showing its interaction with external entities. It will illustrate how students, departmental officers, the Director, the Academic Registrar, and the system administrator interact with the SGCS. It is chosen because it clearly defines the system boundary and the flow of information between the system and its users.

3.5.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) shows how data flows through the proposed system, including the processes, data stores, and external entities involved. It will model processes such as user authentication, clearance requests, approvals, notifications, and report generation. It is chosen because it provides a clear understanding of how information is processed within the system.

3.5.4 System Flowchart

A System Flowchart is a graphical representation of the sequence of operations performed by the system. It will illustrate the complete workflow from student login and clearance request

submission to departmental approvals and final clearance confirmation. It is chosen because it clearly shows the logical flow of system activities.

3.5.5 Use Case Diagram

A Use Case Diagram illustrates the interactions between users and the system. It will identify system actors such as students, departmental officers, the Director, the Academic Registrar, and the administrator, together with the functions they perform. It is chosen because it clearly defines the functional requirements from the users' perspective.

3.5.6 Sequence Diagrams

A Sequence Diagram shows the order of interactions between users and the system during a specific process. It will model key activities such as user login, clearance approval, transcript request, and notification processing. It is chosen because it helps visualize communication between system components over time.

3.6 Methodology for System Implementation

System implementation is the phase where the system design is converted into a working application using appropriate technologies.

3.6.1 Front-End Development

The front-end of the SGCS will be developed using HTML, CSS, and JavaScript. These technologies will create a responsive and user-friendly interface that allows students and staff to access the system on desktops, tablets, and smartphones.

3.6.2 Back-End Development

The back-end will be developed using PHP. It will handle user authentication, process clearance requests, enforce the workflow, manage user access, and communicate with the database.

3.6.3 Database Management

The system will use MySQL as the database management system. It will store and manage student records, clearance requests, approvals, officer information, and other system data while ensuring data integrity and efficient retrieval.

3.7 Methodology for System Testing

System testing is the process of verifying that the system functions correctly, performs reliably, and meets user requirements before deployment.

Testing will be conducted in three stages. First, the developer will perform unit and integration testing to ensure that all system components work correctly together and that any identified errors are corrected.

Second, User Acceptance Testing (UAT) will involve four final-year students and two administrative staff from Tangaza University. Participants will test key system functions such as submitting, approving, and tracking clearance requests. Their feedback will be collected through evaluation forms and used to improve the system before deployment.

Finally, heuristic evaluation will be conducted to assess the system's usability, consistency, and ease of use. Any usability issues identified will be addressed before the system is deployed.

3.8 Methodology for System Deployment

System deployment is the process of making a completed and tested system available for use by its intended users.

The SGCS will be deployed in two phases. Phase One will involve a pilot implementation with the Library and four final-year students to identify and correct any issues. Phase Two will involve full deployment to all six clearance departments and eligible final-year students.

Before deployment, departmental officers will receive training on using the system, while students will have access to a user guide. During the pilot phase, the manual clearance process will remain available as a backup to ensure a smooth transition.

3.9 Chapter Summary

This chapter outlined the methodology for developing the Students' Graduation Clearance System (SGCS). Requirements will be gathered through document analysis and questionnaires. The SSAD methodology will guide system analysis and design using context diagrams, DFDs, flowcharts, ERDs, use cases, sequence diagrams, and pseudocode. The system will be implemented using standard web technologies with a MySQL database. Testing will include developer testing and User Acceptance Testing (UAT), with user feedback used to refine the system before deployment. Finally, the system will be deployed in phases, beginning with a pilot implementation followed by full deployment across all clearance departments.

CHAPTER FOUR: SYSTEM ANALYSIS

4.1 Chapter Introduction

System analysis is the process of examining an existing system to understand how it operates, identify its strengths and weaknesses, and determine the requirements for an improved system. It provides the foundation for designing an effective solution.

This chapter analyses the current manual graduation clearance process at Tangaza University using the SSAD methodology. It evaluates the system's strengths and weaknesses, assesses the feasibility of the proposed Students' Graduation Clearance System (SGCS), defines the functional and non-functional requirements, and presents key analysis models, including the context diagram, data flow diagram (DFD), flowchart, and actor descriptions.

4.2 Description of the Current System

The current graduation clearance process at Tangaza University is fully manual and paper-based. After meeting the academic and financial requirements for graduation, a student collects a graduation clearance form from the Academic Registrar's Office and visits six departments to obtain clearance signatures and stamps.

The departments involved are:

1. Institute Directorate – Confirms that all departmental property has been returned.
2. University Library – Verifies that all borrowed books and library materials have been returned.
3. Dean of Students' Office – Checks the student's conduct record and any outstanding student affairs obligations.
4. Administration and University Store – Confirms that all university-issued items have been returned.
5. Finance Office – Verifies that the student has cleared all tuition fees and financial obligations.
6. Academic Registrar's Office – Performs the final verification and approves the student's eligibility for graduation after confirming clearance from all departments.

Each department signs and stamps the clearance form before the student proceeds to the next stage. The completed form is then submitted to the Academic Registrar, marking the end of the graduation clearance process.

4.3 Strengths of the Current System

1. Simple Process – The clearance procedure is easy to understand and follow, requiring students to collect signatures from the relevant departments.
2. Low Infrastructure Requirements – The system operates without computers, internet access, or specialized software, making it inexpensive to run.
3. Direct Verification – Face-to-face interactions allow staff to verify student records and resolve issues immediately where necessary.
4. Reliable During Technical Failures – Since it is paper-based, the system continues to function during power outages or network disruptions.
5. Familiar to Users – Staff and students are already accustomed to the process, reducing the need for training and making it easy to implement consistently.

4.4 Weaknesses of the Current System

1. Time-Consuming Process – Students must visit multiple offices physically, resulting in long queues and repeated trips before completing clearance.
2. No Real-Time Tracking – Students cannot monitor their clearance progress without visiting the respective offices.
3. Risk of Lost or Damaged Forms – Since the process relies on a single paper form, loss or damage may require students to restart the entire process.
4. Poor Inter-Departmental Communication – Departments communicate manually, causing delays in resolving student clearance issues.
5. Data Duplication and Errors – Student information is recorded repeatedly across departments, increasing the risk of inconsistencies and transcription errors.
6. Lack of Centralized Reporting – The Academic Registrar cannot easily monitor clearance progress, identify bottlenecks, or generate timely reports.
7. Student Inconvenience and Stress – The manual process is cumbersome, requiring multiple campus visits and creating unnecessary anxiety during the graduation period.

4.5 Feasibility Study

Feasibility Study is the process of evaluating whether a proposed system or project is practical and achievable before development begins. It assesses factors such as technical capability, cost, operational suitability, time, and legal requirements to determine whether the project should proceed.

4.5.1 Technical Feasibility

Technical feasibility assesses whether the required technology, infrastructure, and technical expertise are available to successfully develop and implement the proposed system (Satzinger, Jackson, & Burd, 2016).

The proposed SGCS is technically feasible because it uses proven web technologies (HTML, CSS, JavaScript, PHP, and MySQL), which are widely available and easy to maintain. The system can run on standard web hosting without requiring specialized hardware. Tangaza University has adequate ICT infrastructure, internet connectivity, and staff with basic computer skills to support the system.

4.5.2 Economic Feasibility

Economic feasibility evaluates whether the expected benefits of a system outweigh its development and operational costs (Shelly & Rosenblatt, 2012).

The SGCS is economically feasible because it is developed using free and open-source software, keeping implementation costs low (estimated at KES 22,110). The system will reduce expenses associated with printing, paperwork, storage, and manual processing while improving efficiency, resulting in long-term cost savings that outweigh the initial investment.

4.5.3 Operational Feasibility

Operational feasibility determines whether the proposed system will be accepted and effectively used by its intended users (Dennis, Wixom, & Tegarden, 2020).

The SGCS is operationally feasible because students and staff are expected to adopt it easily. Students can submit and track clearance online, while departmental officers can process requests through a simple interface with minimal training.

4.5.4 Schedule Feasibility

Schedule feasibility assesses whether the project can be completed within the available time (Satzinger et al., 2016).

The SGCS is schedule feasible because its scope is manageable and can be completed within the planned seven-month development period. The use of familiar technologies and a well-defined project schedule makes timely completion achievable

4.5.5 Legal and Ethical Feasibility

Legal and ethical feasibility determines whether a system complies with legal requirements and ethical principles governing its operation.

The SGCS complies with the Kenya Data Protection Act, 2019 by safeguarding student data through secure authentication, role-based access control, and secure data transmission.

The system promotes fairness, transparency, and equal treatment by applying the same clearance process to all students without discrimination while respecting users' privacy and cultural diversity.

4.5.6 Feasibility Study Conclusion

The feasibility study confirms that the proposed Students' Graduation Clearance System is technically, economically, operationally, schedule, and legally feasible. The required resources are available, the benefits outweigh the costs, the project can be completed within the planned timeframe, and the system complies with relevant legal and ethical standards. Therefore, the SGCS is suitable for implementation at Tangaza University.

4.6 Requirements Analysis

The requirements analysis presented in this section is the product of the document analysis and questionnaire activities described in Chapter Three, supplemented by a thorough examination of the five related systems reviewed in Chapter Two. It establishes, in precise and verifiable terms, what the proposed SGCS must do, the standards it must meet, and the data it must manage. Requirements have been categorised into three groups: functional requirements, which specify the capabilities the system must provide; non-functional requirements, which define the performance and quality standards it must achieve; and data requirements, which identify the information the system must store and manage.

4.6.1 Functional Requirements

Functional requirements describe what the system will do - the specific functions and services it will provide to users.

1. The system will allow users (students, officers, Director, Registrar, and Administrator) to log in securely.
2. The system will allow students to manage their profiles.
3. The system will allow students to initiate graduation clearance requests.
4. The system will allow departmental officers to approve or reject clearance requests and provide comments for rejections.
5. The system will allow students to track their clearance status in real time.
6. The system will send notifications for submissions, approvals, rejections, and final clearance.
7. The system will allow the Finance Office to verify financial clearance and record outstanding balances.
8. The system will provide reporting dashboards for the Director and Academic Registrar.
9. The system will maintain an audit trail of all system activities.

4.6.2 Non-Functional Requirements

Non-functional requirements describe how the system will perform, focusing on quality attributes such as security, reliability, and usability.

1. **Security:** The system protects user data through secure authentication and role-based access control.
2. **Reliability:** The system will operate consistently without data loss and support regular data backups.
3. **Availability:** The system will be accessible during normal university working hours with minimal downtime.
4. **Usability:** The system will provide a simple and user-friendly interface that is easy to learn and use.
5. **Performance:** The system will load pages quickly and support multiple users simultaneously without significant delays.
6. **Scalability:** The system will accommodate future growth in users and records without major modifications.
7. **Maintainability:** The system will be easy to maintain, update, and enhance through well-organized and documented code.
8. **Responsiveness:** The system will function effectively on smartphones, tablets, and desktop computers.

4.6.3 Data Requirements

Table 4.6.3 Data Requirements

Data Category	Description	Key Requirements
Student Data	Stores student details such as registration number, name, programme, institute and login credentials.	Registration must be unique; required fields cannot be empty.
Clearance Records	Stores clearance requests, current workflow stage, and clearance status.	Linked to the corresponding student and updated throughout the clearance process.
Departmental Approval Records	Stores approval or rejection details from each department.	Includes department, officer, action taken, transcript, timestamp and comments.
Officer Data	Stores details of departmental officers and administrators.	Includes names, department, role, login credentials and contact information.
Notification Records	Stores all system generated notifications sent to users.	Linked to the respective student and accessible only to authorized users.

4.7 Current System Users

Table 4.7 Current System Users

Actors	Role in the current system
Student	Completes the graduation application form, submits it to the Registrar, obtains the clearance form, visits each department for approval, and returns the completed form to the Registrar.
Academic Programme Leader	Verifies whether the student is eligible for graduation, signs the graduation application form, and forwards it to the Registrar's Office.
Institute Director	Confirms that the student has no outstanding institute materials and signs the clearance form.

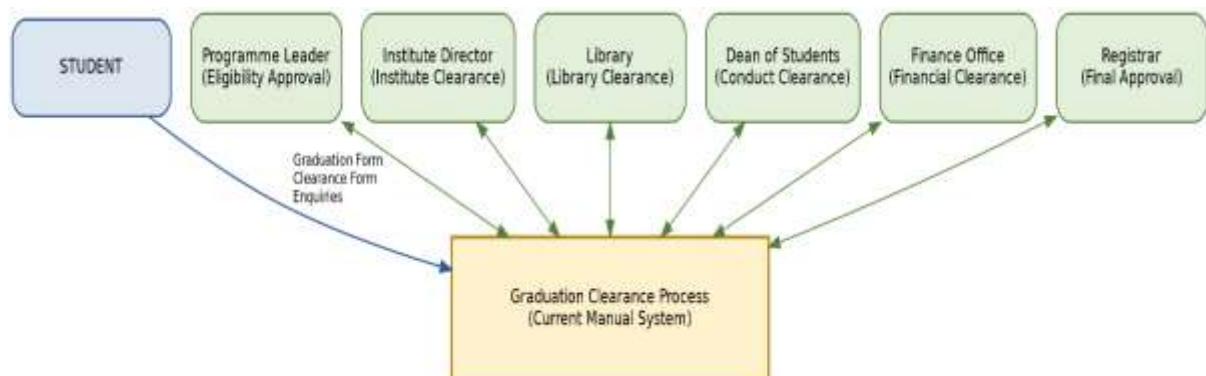
Librarian	Verifies that all library books and materials have been returned before approving the student's clearance.
Dean of Students	Confirms the student's conduct status and approves clearance where there are no outstanding issues.
Administration/University Store	Verifies that all university property has been returned or records any outstanding items before approving clearance.
Finance Officer	Confirms that the student has cleared all financial obligations or records any outstanding balances before approving clearance.
Academic Registrar	Reviews the completed clearance form, confirms that all departmental approvals have been obtained and grants the final graduation clearance

4.8 Process logic Design

4.8.1 Context Diagram

The context diagram provides a high-level view of the current manual graduation clearance process at Tangaza University. It identifies the external entities that interact with the system, namely the Student, Academic Programme Leader, Institute Director, Librarian, Dean of Students, Finance Officer, and Academic Registrar. The diagram illustrates the flow of information between these actors and the graduation clearance process, defining the system boundary without showing internal processing activities.

Figure 4.8.1: Context Diagram - Current Manual Clearance System

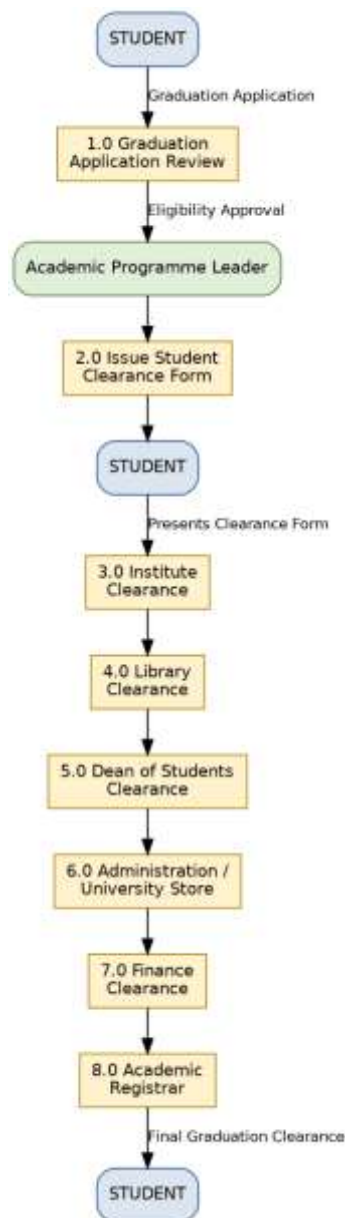


4.8.2 Level 1 DFD

The Level 1 Data Flow Diagram (DFD) illustrates how data flows through the current manual graduation clearance process. It decomposes the overall process into its major activities,

beginning with graduation application approval, followed by sequential clearance through the Institute Director, Library, Dean of Students, Administration/University Store, Finance Office, and finally the Academic Registrar. The DFD shows how information moves between students and departments and highlights the sequential nature of the clearance process.

Figure 4.8.2 Level 1 DFD



4.8.3 Inputs and Outputs

Table 4.8.3 Input and Outputs

Category	Description
Inputs	Graduation application form, student details (registration number, programme, contact

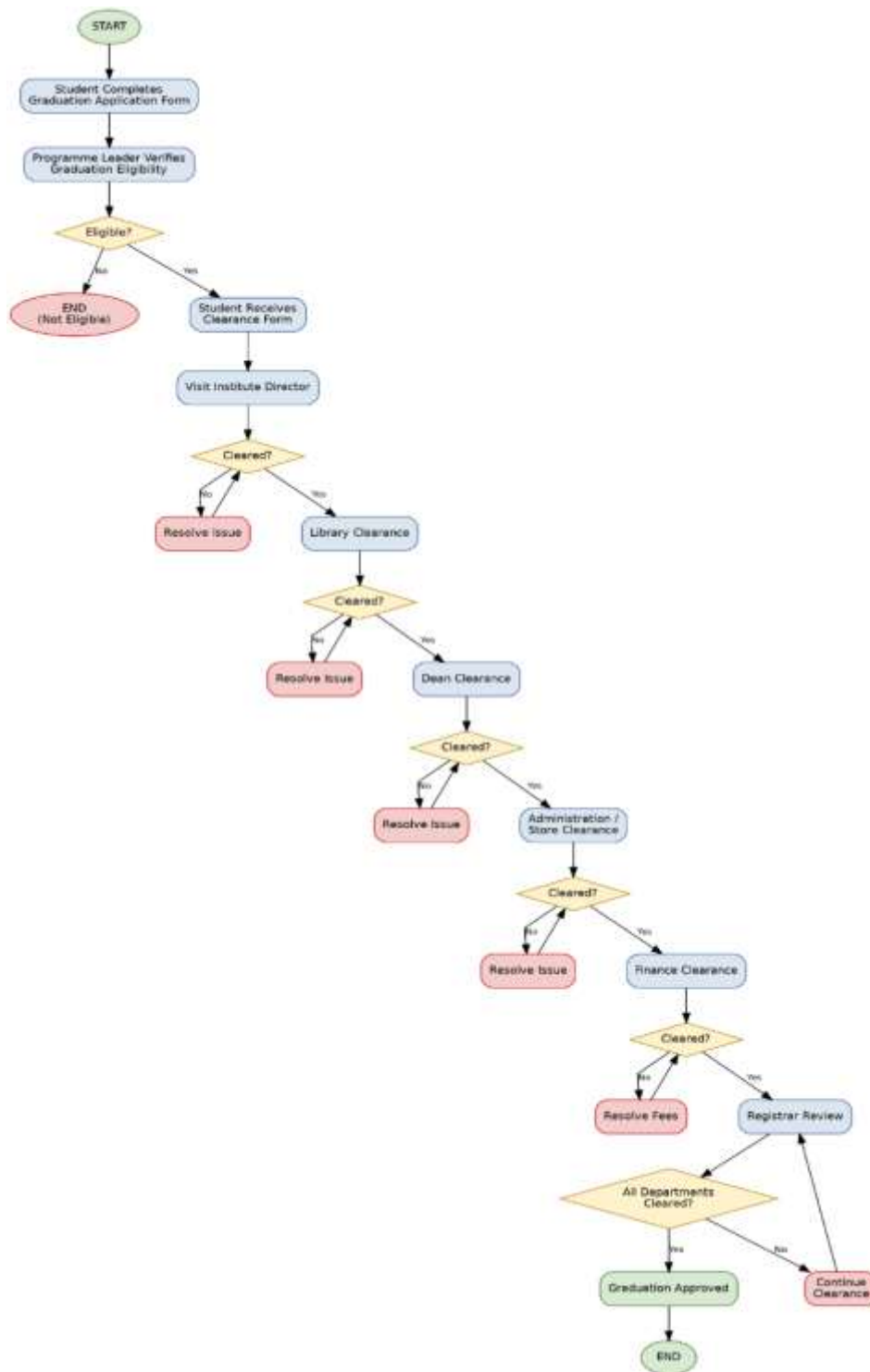
	information), physical clearance form, departmental approval/rejection decisions, comments, signatures, stamps, and financial clearance details.
Outputs	Updated clearance form showing departmental clearance status, approval or rejection feedback, completed clearance form with all required signatures, and final graduation clearance approval issued by the Academic Registrar.

4.8.4 System Flowcharts

The system flowchart illustrates the step-by-step workflow of the current manual graduation clearance process. It shows the sequence of activities from graduation application, eligibility verification, and departmental clearances to the final approval by the Academic Registrar.

The flowchart also includes decision points where a student may be required to resolve outstanding issues before proceeding to the next department. This representation provides a clear understanding of the operational flow of the current system and identifies where delays may occur.

Figure 4.8.4 System Flowchart



4.8 Chapter Summary

This chapter analyzed the current manual graduation clearance process at Tangaza University, highlighting its strengths, weaknesses, and feasibility. It established that the proposed SGCS is technically, economically, operationally, legally, and schedule feasible. The chapter also defined the system's functional and non-functional requirements, data requirements, and presented key analysis artefacts, including the context diagram, Level 1

DFD, system flowchart, actor descriptions, and input/output specifications, providing the foundation for the system design in the next chapter.

CHAPTER FIVE: SYSTEM DESIGN

5.1 Introduction

System design is the process of translating the system requirements identified during analysis into a blueprint for developing the proposed solution. It defines how the system will be structured, how data will be organized and processed, and how users will interact with the system.

5.2 Description of the Proposed System

The proposed Students' Graduation Clearance System (SGCS) is a web-based system designed to automate and centralize the graduation clearance process at Tangaza University. The system provides role-based access for students, departmental officers, the Director, the Academic Registrar, and the System Administrator through secure login credentials.

Students can initiate graduation clearance online, track the status of their clearance in real time, receive in-app notifications, upload required supporting documents, and view their final approval document once all clearance stages are successfully completed. The Academic Registrar uploads transcripts. Departmental officers review, approve, or reject clearance requests through dedicated dashboards, while the system automatically updates records and forwards approved requests to the next department in the workflow.

The Director and Academic Registrar will access reporting dashboards to monitor clearance progress and generate reports, while the System Administrator will manage user accounts, departments, and system settings. By replacing the manual paper-based process with an automated workflow, the SGCS will improve efficiency, transparency, communication, and data management throughout the graduation clearance process.

5.2.1 Strengths of the Proposed System

1. Automates the entire graduation clearance workflow, reducing paperwork and processing time.
2. Provides real-time clearance status tracking and automated notifications for students and officers.
3. Centralizes student clearance records, improving data accuracy and accountability.

4. Generates management reports to support monitoring and decision-making.
5. Supports role-based access control to enhance data security and privacy.

5.2.2 Weaknesses of the Proposed System

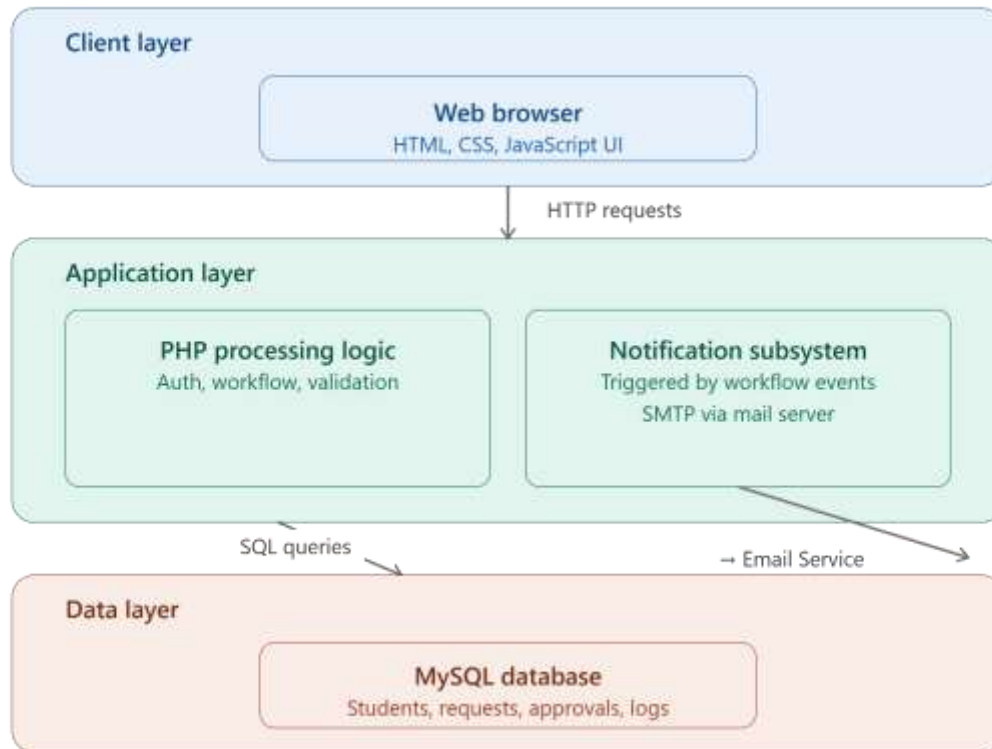
1. Depends on reliable internet connectivity for system access.
2. Requires initial training for departmental officers and system users.
3. May experience service interruptions if the hosting server is unavailable.
4. Is exposed to cybersecurity threats if appropriate security measures are not maintained.
5. Some users may initially resist transitioning from the manual process to the digital system.

5.5 Conceptual Architecture of the Proposed System

The SGCS adopts a three-tier client-server architecture consisting of the presentation layer, application layer, and data layer. This architecture separates user interaction, business logic, and data storage, making the system easier to develop, maintain, and scale.

- **Presentation Layer:** Provides the user interface through a web browser on desktops, laptops, tablets, and smartphones. It is developed using HTML, CSS, and JavaScript, allowing users to interact with the system through responsive web pages.
- **Application Layer:** Implements the system's business logic using PHP. It processes user requests, manages the sequential clearance workflow, validates data, generates notifications, and communicates with the database.
- **Data Layer:** Uses a MySQL relational database to store student records, user accounts, clearance requests, approval records, notifications, and graduation approval documents. The database ensures data integrity through relationships and constraints, while supporting efficient data retrieval and updates.

Figure 5.5 Conceptual Architecture



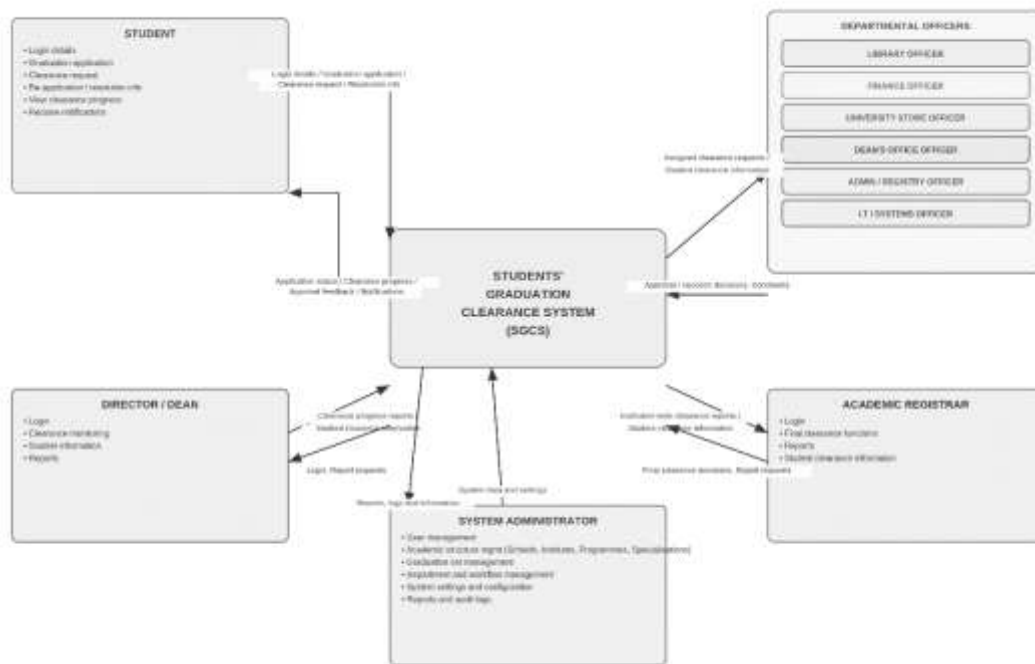
5.6 Process Design Using Appropriate Tools

5.6.1 Context Diagram

The proposed Context Diagram presents the SGCS as a single system and illustrates how it interacts with seven external entities: Student, Director, Library, Finance, Dean, Academic Registrar, and System Administrator. It shows the major data exchanged between the users and the system, including login details, clearance requests, approvals, notifications, reports, documents, and graduation approval documents. The context diagram provides a high-level view of the system boundary and its external interactions without showing internal processes or data stores.

Figure 5.6.1 Context Diagram (Proposed System)

Figure 5.6.1 Context Diagram of the Proposed SGCS



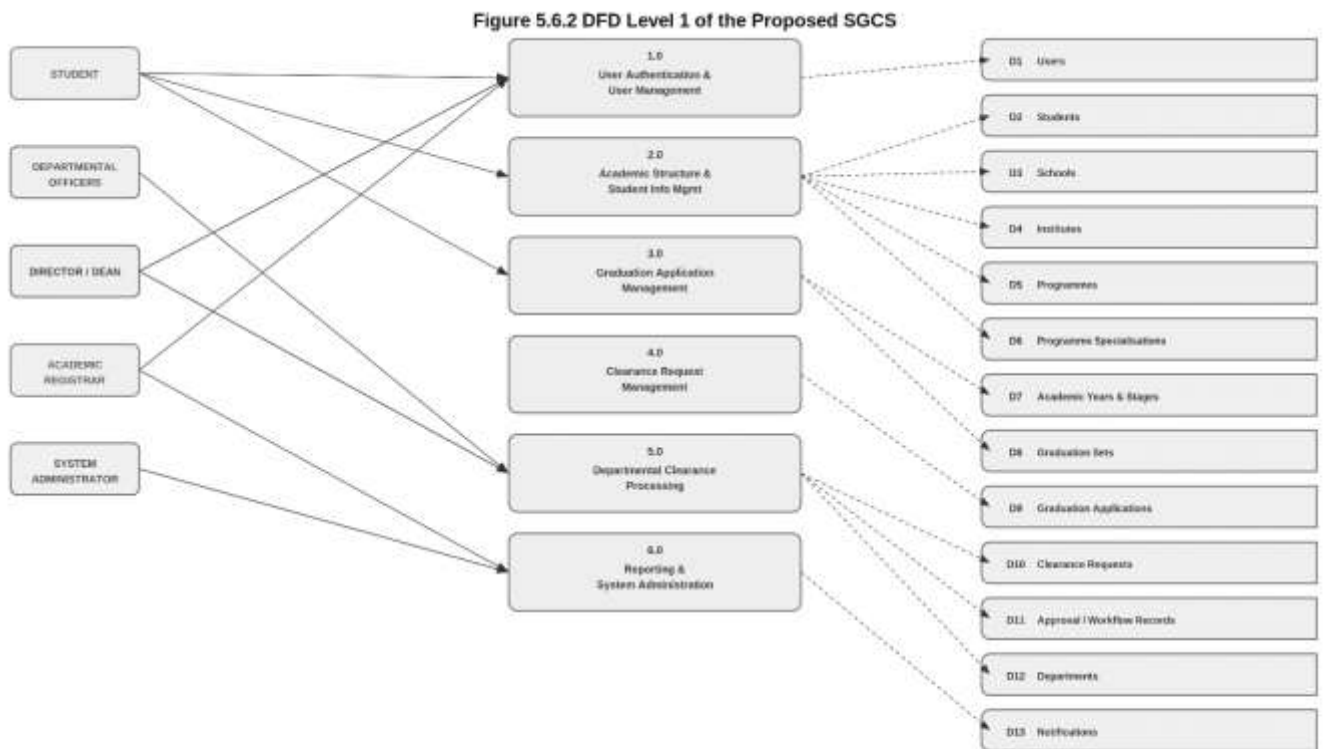
5.6.2 DFD Level 1

The Level 1 Data Flow Diagram (DFD) decomposes the SGCS into its major functional processes and illustrates how information flows between users, system processes, and the database. The proposed SGCS consists of the following major processes:

- **User Authentication and Account Management** - This process manages user login and access according to assigned roles. The system controls access to features based on roles such as Student, Officer (Library, Finance, University store), Director, Dean, Academic Registrar, and Administrator.
- **Academic Information Management** - This process manages the student's academic information, including School, Institute, and Programme. These academic records are maintained through the administrative functions and are used to provide accurate student information throughout the system.
- **Graduation Set Management** - Administrators manage graduation sets, and students are assigned to a graduation set. Clearance can be initiated only when the assigned graduation set is active.
- **Clearance Request Management** - After satisfying the graduation application requirements, the student can initiate the clearance process. The system creates a clearance request and tracks its overall progress throughout the departmental workflow.

- Departmental Clearance Processing - Each department reviews the clearance request according to the configured workflow sequence. Officers can approve or reject requests and provide comments. Approved requests progress through the workflow, while rejected requests are returned for resolution before the clearance process can continue.
- Reporting and Administration - This process allows authorised users to access reports and manage system information. Administrators manage academic structures, users, departments, graduation sets, and workflow settings. Directors and other authorised officers can access reports relevant to their responsibilities.

Figure 5.6.2 DFD Level 1(proposed system)



5.6.3 DFD Level 2

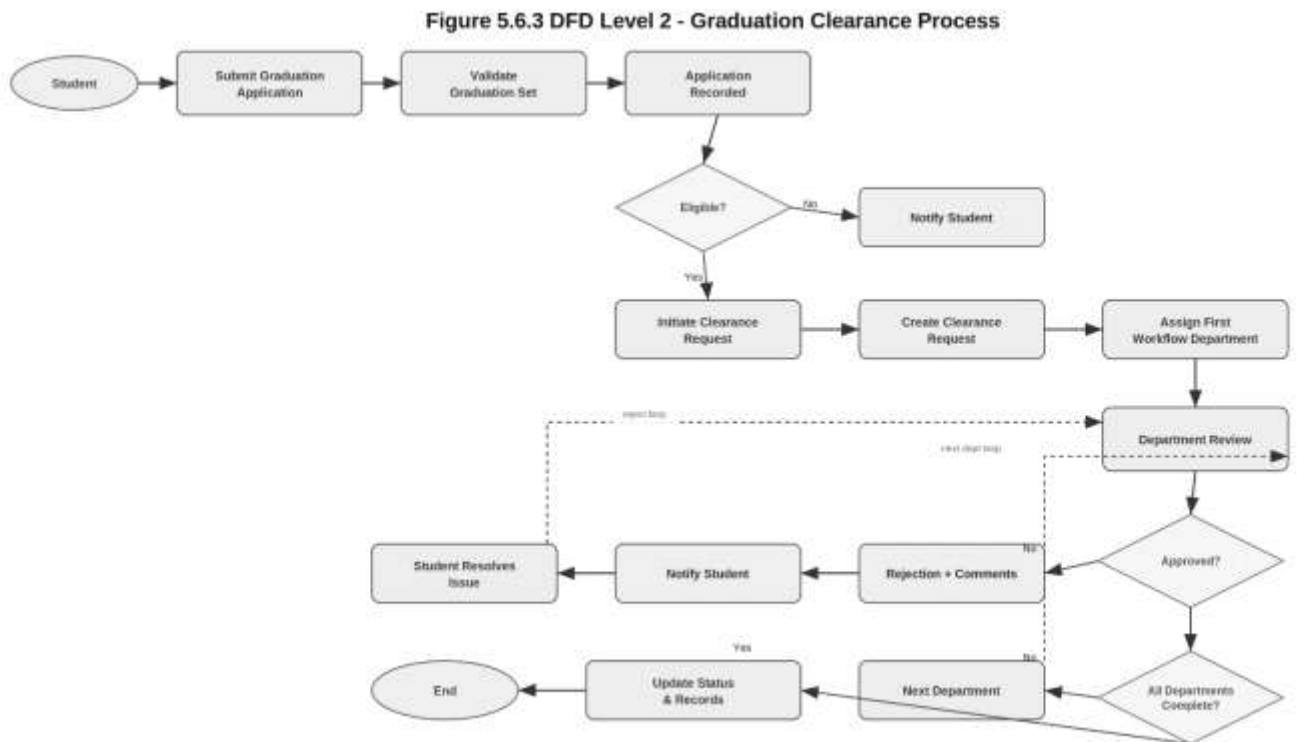
The Level 2 Data Flow Diagram expands the Clearance Request Management process to provide a more detailed view of the clearance workflow.

The system performs the following activities:

- Receives the student's clearance request.
- Validates the student's eligibility.
- Creates a clearance request record.
- Creates or activates the appropriate departmental workflow records.
- Records either an approval or rejection decision.
- Advances the request when the required workflow conditions are satisfied.
- Generates notifications to relevant users.
- Completes the clearance process after all required approvals and final review are completed.

The Level 2 DFD provides a detailed representation of how information moves through the clearance workflow.

Figure 5.6.3 DFD Level 2



5.6.4 Use Case Diagram

The Use Case Diagram illustrates how the different users interact with the proposed Students' Graduation Clearance System. The main actors in the system are the Student, Departmental Officer, Director/Dean, Academic Registrar, and System Administrator.

Student:

- Register or access their account
- Log into the system
- View academic information
- Initiate a clearance request
- Monitor clearance progress
- View approval and rejection feedback
- Resolve rejected clearance requirements
- View notifications

Departmental Officer (Librarian, Finance, University store):

- Log into the system
- View assigned clearance requests
- Review student requests
- Approve requests
- Reject requests
- Provide comments
- Process requests according to departmental responsibilities

Director/Dean:

- Log into the system
- Access clearance progress information
- Monitor students within the relevant academic structure
- View clearance reports
- Perform assigned clearance responsibilities where applicable

Academic Registrar:

- Log into the system
- Monitor institutional clearance progress
- View clearance reports
- Access student clearance information
- Perform final clearance-related responsibilities

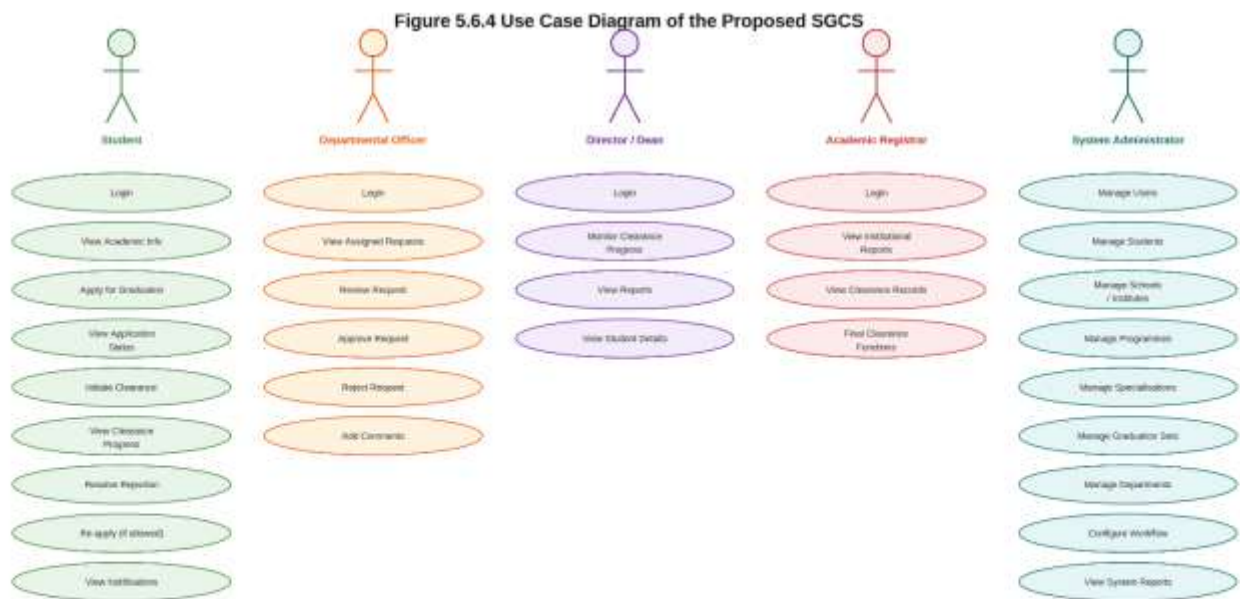
System Administrator:

- Manage users
- Manage user roles

- Manage student information
- Manage graduation sets
- Manage administrative departments
- Configure workflow order
- Access administrative reports

The Use Case Diagram provides an overview of the system functions available to each user according to their assigned role.

Figure 5.6.4 Use Case Diagram



5.6.5 System Flowcharts

System flowcharts are used to represent the major processes within the SGCS in a step-by-step manner. The proposed system includes separate flowcharts for the major users and processes.

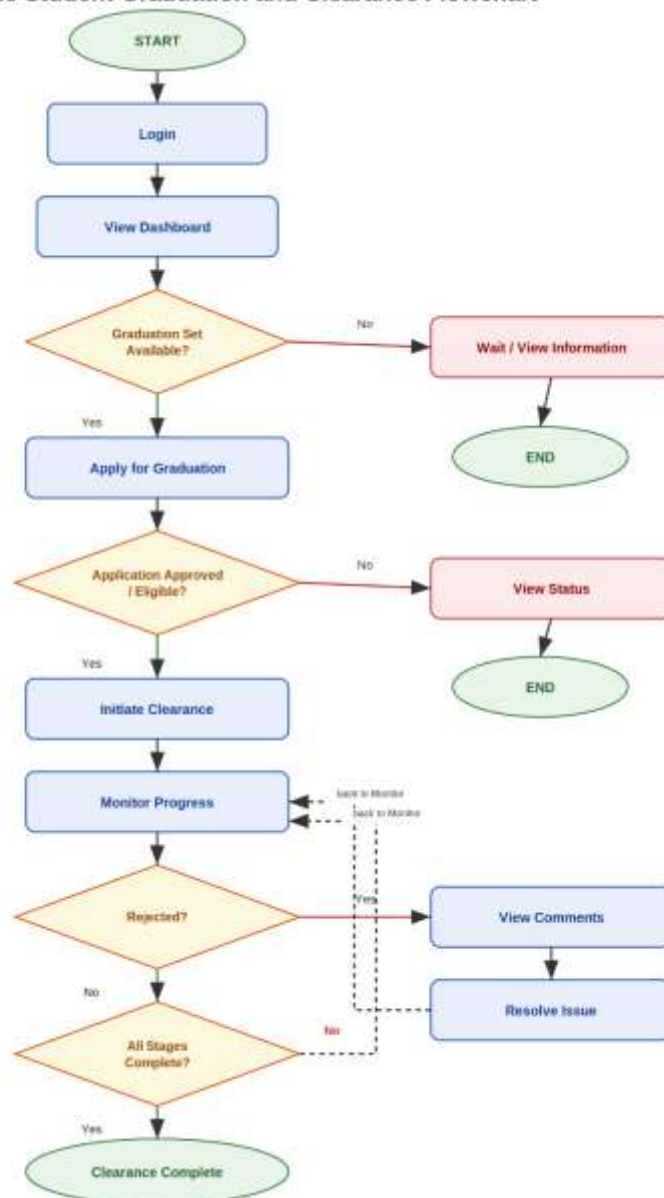
Student Clearance Flowchart: The Student Clearance Flowchart represents the process followed by a student. The process is as follows:

- Student logs into the SGCS.
- Student accesses their academic information.
- Student is assigned to an active graduation set.

- The system validates the clearance-eligibility requirements.
- The student initiates the clearance process when eligible.
- The system creates the clearance request.
- The request is processed according to the departmental workflow.
- The student monitors the clearance progress.
- If a department rejects the request, the student views the comments.
- The student resolves the identified issue.
- The clearance request continues according to the system workflow.
- Once all required approvals are completed, the final clearance status is updated.

Figure 5.6.5 Students Flow Chart

Figure 5.6.5 Student Graduation and Clearance Flowchart



Departmental Officer Workflow - The Departmental Officer Flowchart illustrates how officers process requests assigned to their department.

The process is as follows:

- Officer logs into the SGCS.
- Officer views assigned clearance requests.
- Officer selects a request for review.
- Officer reviews the student's clearance information.
- Officer makes a decision.

If the request is approved:

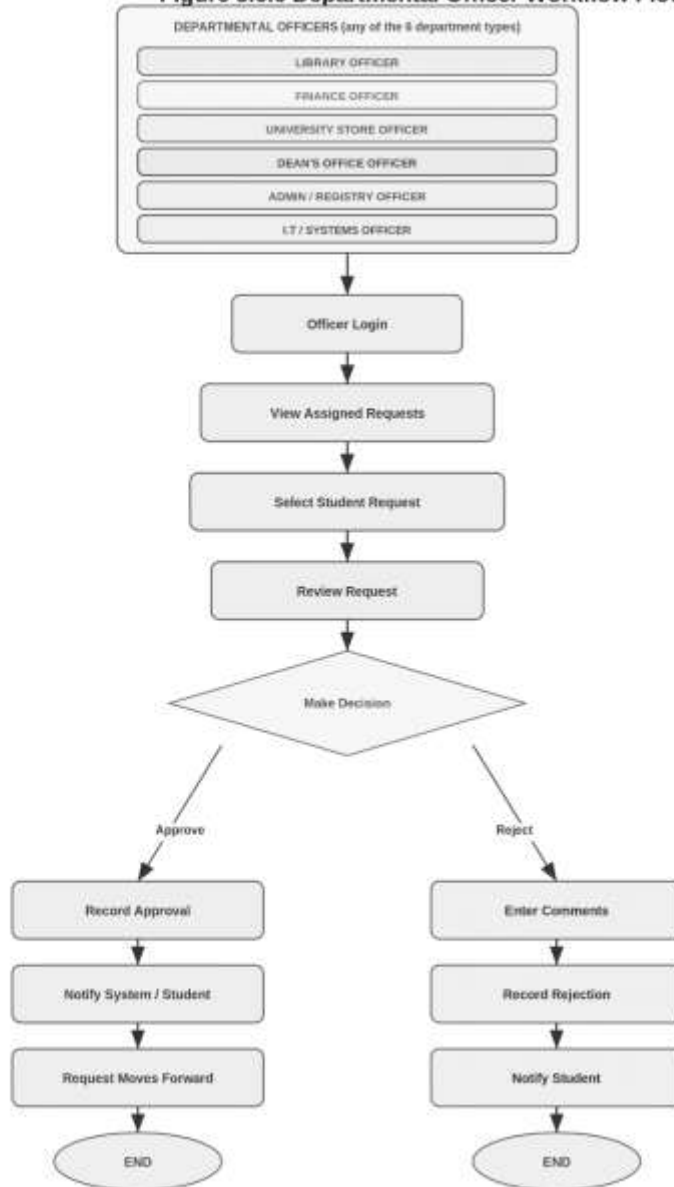
- The approval is recorded.
- The system updates the workflow.
- The request proceeds according to the next configured workflow stage.

If the request is rejected:

- The rejection is recorded.
- The officer provides comments.
- The student is notified.
- The request remains unresolved until the required issue is addressed.

Figure 5.6.5 Departments Flow Chart

Figure 5.6.6 Departmental Officer Workflow Flowchart



5.6.6 Activity Diagram

An Activity Diagram is a UML behavioural diagram used to illustrate the sequence of activities, decisions, and workflow within a system.

The process begins when the student logs into the system. The system validates the student's profile, assigned active graduation set, required supporting documents, and final-approval name confirmation. Once the required conditions are satisfied, the student initiates a clearance request. The system creates the clearance request and processes it according to the configured departmental workflow.

Each responsible officer reviews the request and makes one of two decisions:

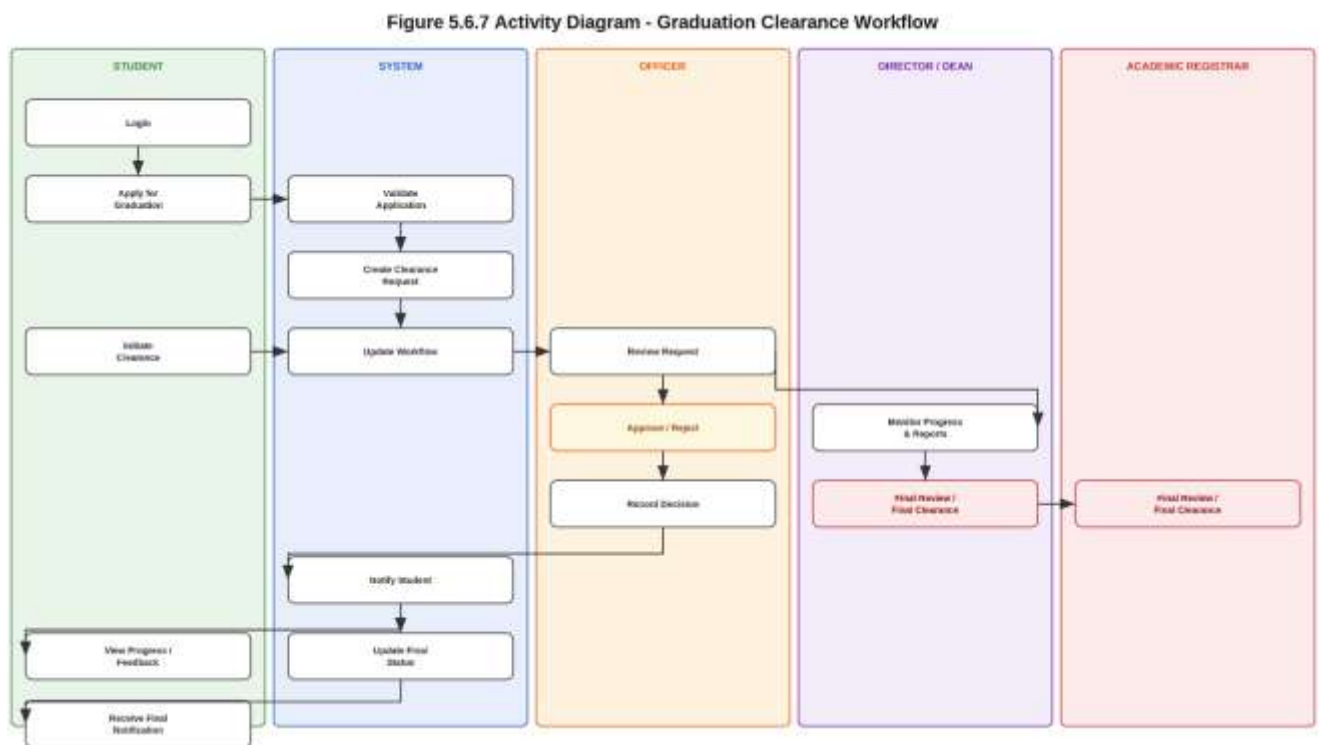
- Approve the request
- Reject the request with comments

When approved, the system updates the workflow and allows the request to continue to the next required stage. When rejected, the system notifies the student and provides the relevant feedback. The student resolves the issue before the clearance process continues.

After the required departmental processing is completed, the appropriate final clearance process is performed and the student's clearance status is updated.

The Activity Diagram was selected because it clearly illustrates the sequence of activities, decision points, and interaction between users and the system.

Figure 5.6.6 Activity Diagram



5.6.7 Sequence Diagrams

A Sequence Diagram is a UML interaction diagram that illustrates communication between users and system components over time.

For the proposed SGCS, sequence diagrams are used to illustrate important processes.

Clearance Initiation Sequence Diagram - The participants include:

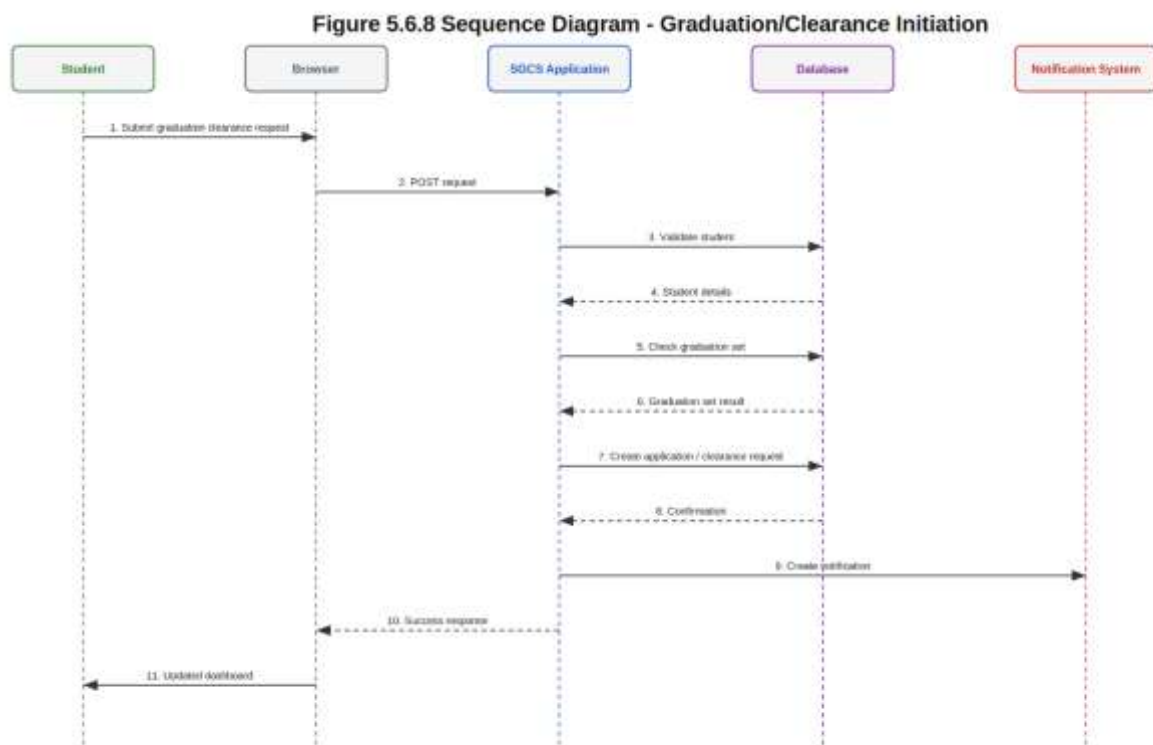
- Student
- Web Browser
- SGCS Application

- Database

The process occurs as follows:

- The student initiates a clearance request.
- The browser sends the request to the SGCS application.
- The application validates the student's eligibility.
- The application checks the relevant graduation-set assignment, profile, and supporting-document information.
- The application communicates with the database.
- The database returns the required validation information.
- The application creates the clearance request.
- The appropriate workflow records are created.
- The system generates relevant notifications.
- A successful response is returned to the student's browser.
- The student's dashboard displays the updated clearance information.

Figure 5.6.7 Sequence Diagram – Clearance Initiation



Departmental Approval Sequence Diagram - The Departmental Approval Sequence Diagram illustrates communication during the review of a clearance request.

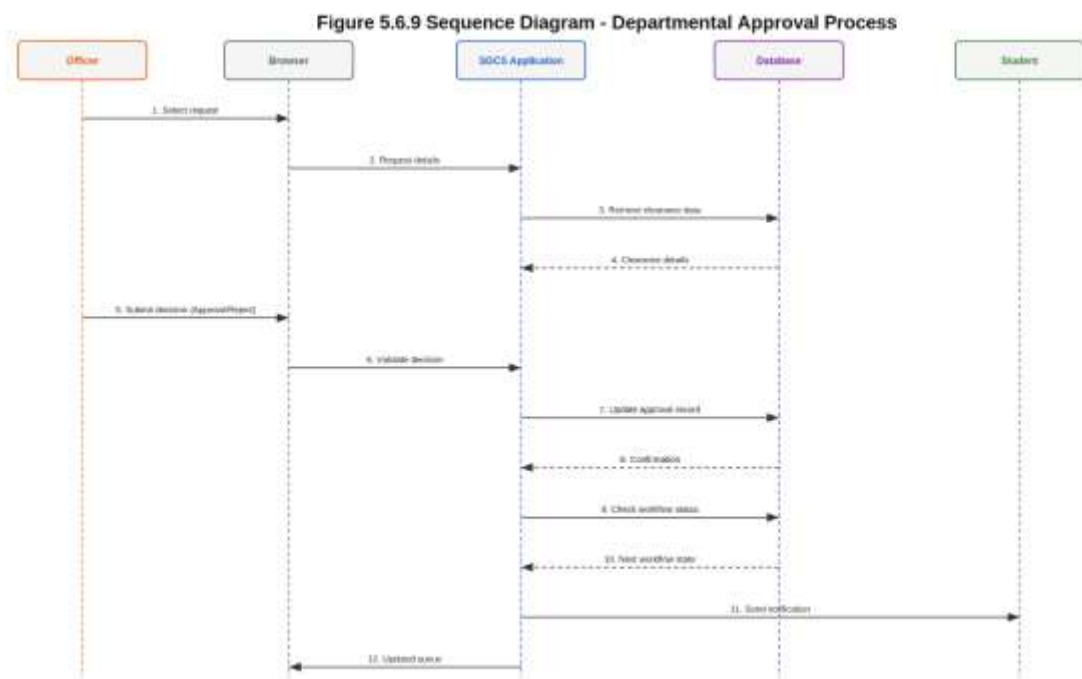
The participants include:

- Departmental Officer
- Web Browser
- SGCS Application
- Database

The process occurs as follows:

- The officer selects a clearance request.
- The browser sends the request to the application.
- The application retrieves the relevant clearance information.
- The officer submits an approval or rejection decision.
- The application validates the submitted decision.
- The application updates the relevant clearance record in the database.
- The system checks the current workflow status.
- The application determines the next required action.
- Notifications are generated where necessary.
- The application returns the updated result to the officer's browser.

Figure 5.6.7 Sequence Diagram – Departmental Approval



5.7 Database Design

The database design for the SGCS has been developed through the entity-relationship modelling process described in Section 3.5 and reflects the data requirements identified in Section 4.6.3. The design prioritises referential integrity, minimal redundancy, and efficient query performance for the access patterns that the application will generate most frequently.

5.7.1 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the major entities in the SGCS and the relationships between them. The current SGCS database includes entities related to the following areas:

User Management -The Users entity manages:

- User accounts
- Usernames
- Password information
- Roles

Student Management - The Students entity stores student personal information and academic assignments.

A student is associated with the following academic information:

- School
- Institute
- Programme

Academic Structure - The academic structure is organised through related entities including:

- Schools
- Institutes
- Programmes
- Programme Specialisations

Graduation Management - Graduation Sets manage graduation periods and related academic information. Students are assigned to a graduation set, and clearance initiation requires that assigned set to be active.

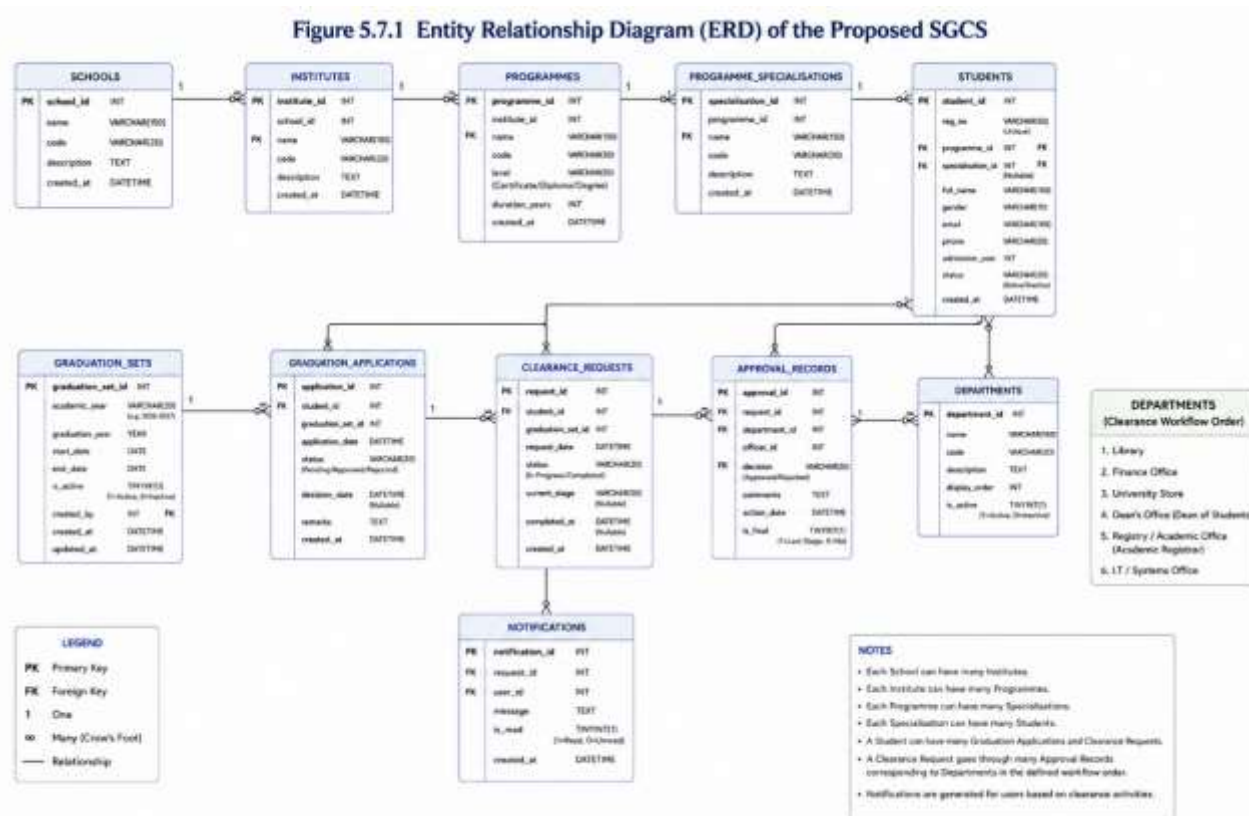
Clearance Management - Clearance Requests record student clearance applications and their overall progress. Related clearance records track departmental processing and decisions.

Department Management - Departments are maintained separately and include workflow sequence information. This allows the system administrator to control the order in which clearance processing occurs.

Notifications - Notifications record messages generated by the system for users during important events such as approvals, rejections, and workflow updates.

The ERD demonstrates the relationships between these entities and supports the implementation of the SGCS workflow.

Figure 5.7.1 Entity Relationship Diagram



5.7.2 Data Dictionary

The data dictionary below describes each table in the SGCS database, listing the field name, data type, constraints, and purpose.

User and Student Management

Table 5.7.2.1 User and Student Management

Table	Purpose
Users	Stores login accounts, roles and authentication information

Student	Stores student personal and academic information.
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Table 5.7.2.2 Academic structure

Table	Purpose
Schools	Stores the schools within the institution.
Institute	Stores institutes and their academic relationships.
Programmes	Stores academic programmes.

Table 5.7.2.3 Graduation Management

Table	Purpose
Graduation sets	Stores graduation periods, academic years and relevant application dates.
Graduation application tables	Stores student graduation applications, where implemented.

Table 5.7.2.4 Clearance Management

Table	Purpose
Clearance request	Stores student clearance requests and overall status.
Approval_records	Stores individual departmental processing decisions.
departments	Stores administrative clearance departments.
Final approval records	Stores final clearance approval information, where applicable.

Table 5.2.7.5 Support tables

Table	Purpose
notifications	Stores notifications generated for users
documents	Stores student academic documents where implemented.
audit log	Stores administrative or system activity where implemented.

5.7.3 Normalisation

Database normalisation is the process of organising a relational database to reduce unnecessary duplication and improve data integrity.

The SGCS database is designed using normalisation principles to support efficient storage and retrieval of information. The SGCS satisfies First Normal Form because each table stores atomic values and does not contain repeating groups.

- Schools are stored separately from institutes.
- Institutes are stored separately from programmes.

This ensures that each field contains a single meaningful value.

The SGCS satisfies Second Normal Form because non-key attributes depend on the complete primary key of their respective tables. For example, information about a programme is stored in the Programmes table rather than being repeatedly stored within student or clearance records.

The SGCS is designed to satisfy Third Normal Form by ensuring that non-key attributes depend on the primary key and not indirectly on other non-key attributes.

- School information is stored in the Schools table.
- Institute information is stored in the Institutes table.

This approach reduces data redundancy and helps prevent update, insertion, and deletion anomalies. Third Normal Form was selected because it provides an appropriate balance between data integrity, reduced redundancy, maintainability, and efficient database operations for the SGCS.

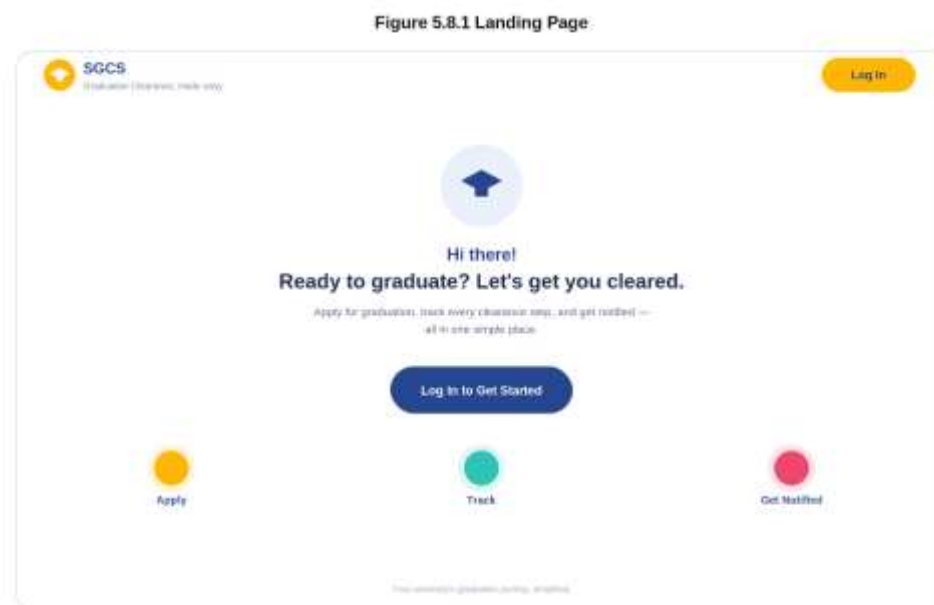
5.8 Input and Output Design

The interface design for the SGCS has been guided by the usability and responsiveness requirements specified in Section 4.6.2 and informed by the review of related systems in Chapter Two, particularly the identification of gaps in mobile responsiveness and notification clarity. Each screen has been prototyped with attention to layout clarity, task focus, and minimal cognitive load for users who may not be highly experienced with web applications.

5.8.1 Landing Page

The landing page is the first interface users see when accessing the SGCS. Its purpose is to introduce the system, establish institutional credibility, and provide access to the login page. It displays the system title, and a brief description of the system, with a positioned Login button serving as the primary call to action.

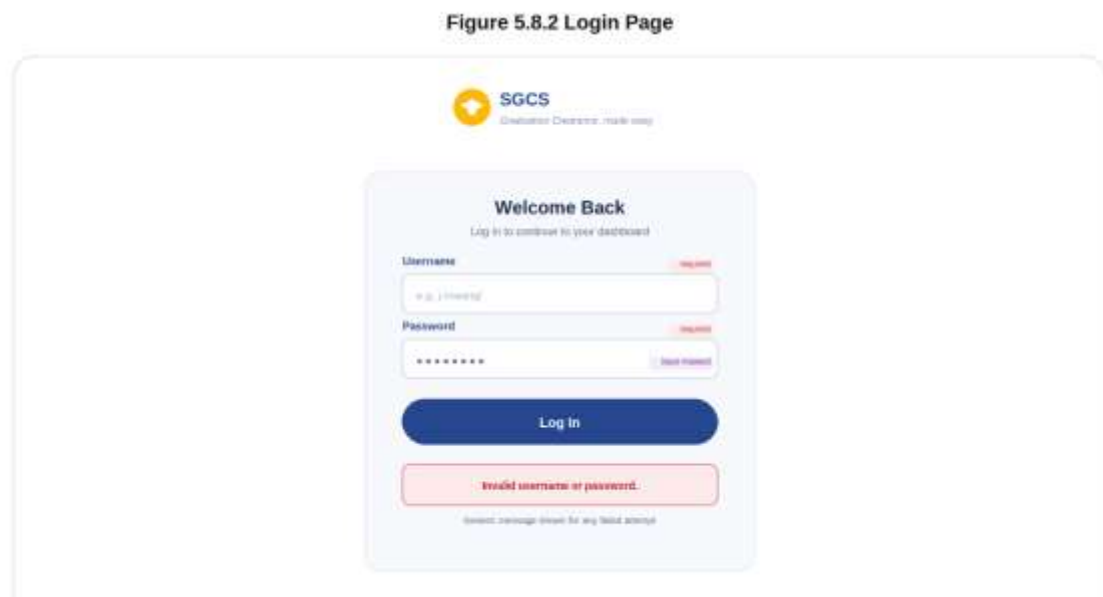
Figure 5.8.1 Landing Page



5.8.2 Login Page

The login page provides secure access to the SGCS through an interface containing Username and Password fields. Input validation ensures that required fields are completed before submission, while the password field masks user input for security. Invalid login attempts display a generic error message to prevent username enumeration. The interface maintains consistent branding with the landing page and is fully responsive across different devices.

Figure 5.8.2 Login Page

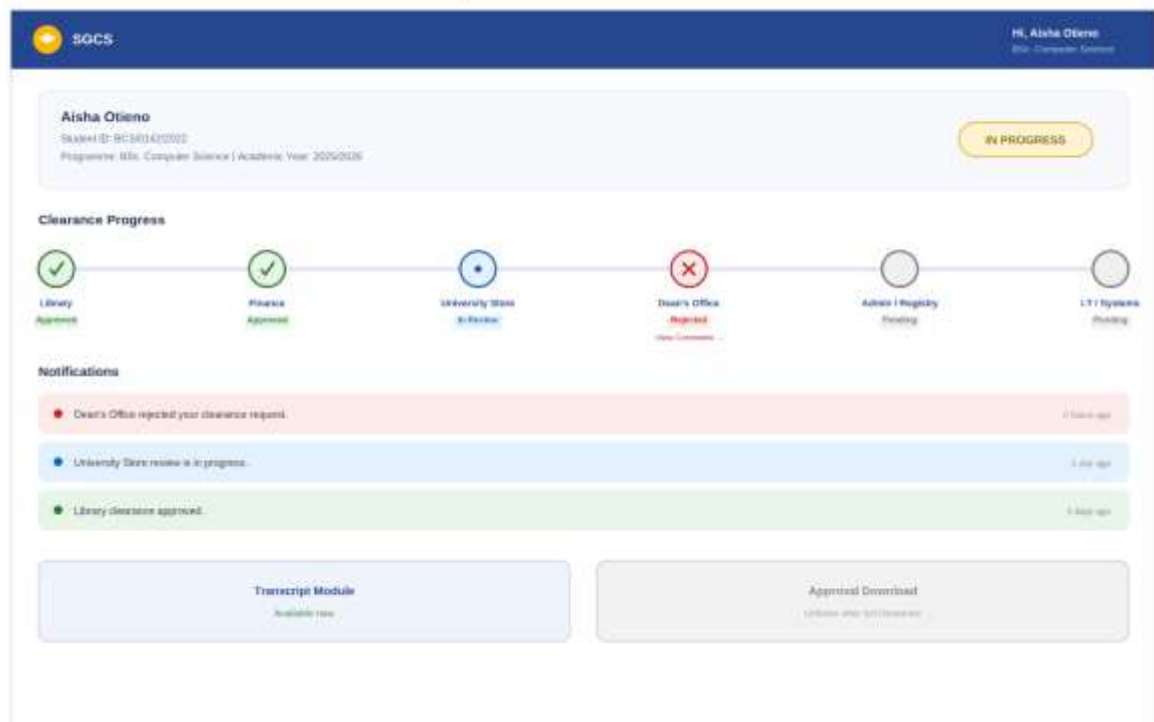


5.8.3 Student Dashboard

The student dashboard serves as the main interface for managing the graduation clearance process. It displays profile-completeness and clearance-eligibility indicators, the current stage, and department progress with status indicators. Rejected stages allow students to view officers' comments. It also provides access to required supporting documents, progress tracking, and the final approval page when available.

Figure 5.8.3 Student Dashboard

Figure 5.8.3 Student Dashboard

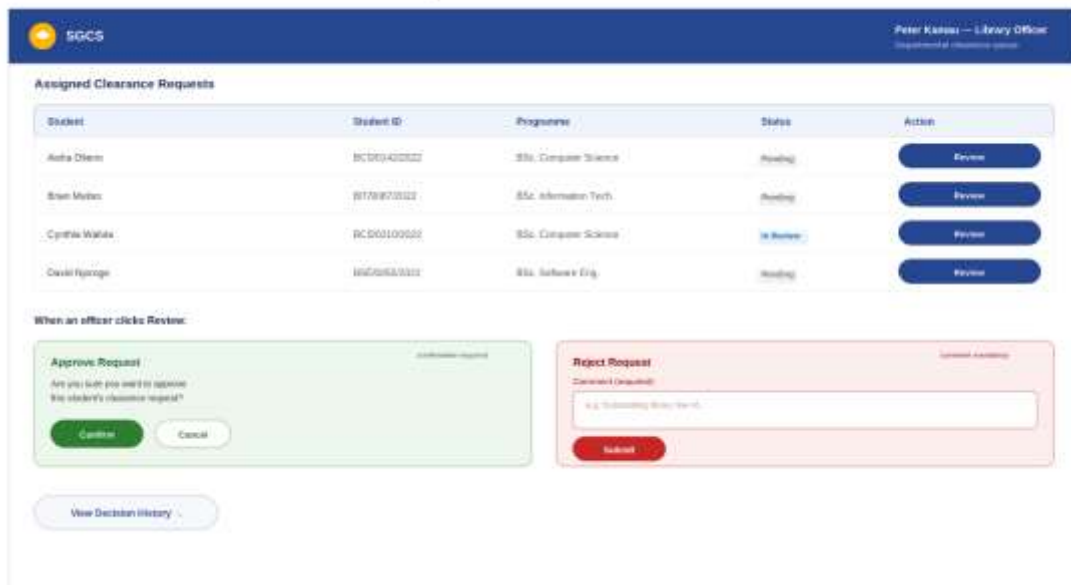


5.8.4 Officer Dashboard

The officer dashboard is designed to support efficient processing of clearance requests. It displays a table of requests assigned to the officer's department, showing the student's details and current status. Officers can Approve or Reject requests through the Request button. Rejections require a mandatory comment before submission, while approvals require confirmation to prevent accidental actions. The dashboard also provides access to the history of previous decisions.

Figure 5.8.4 Officer Dashboard

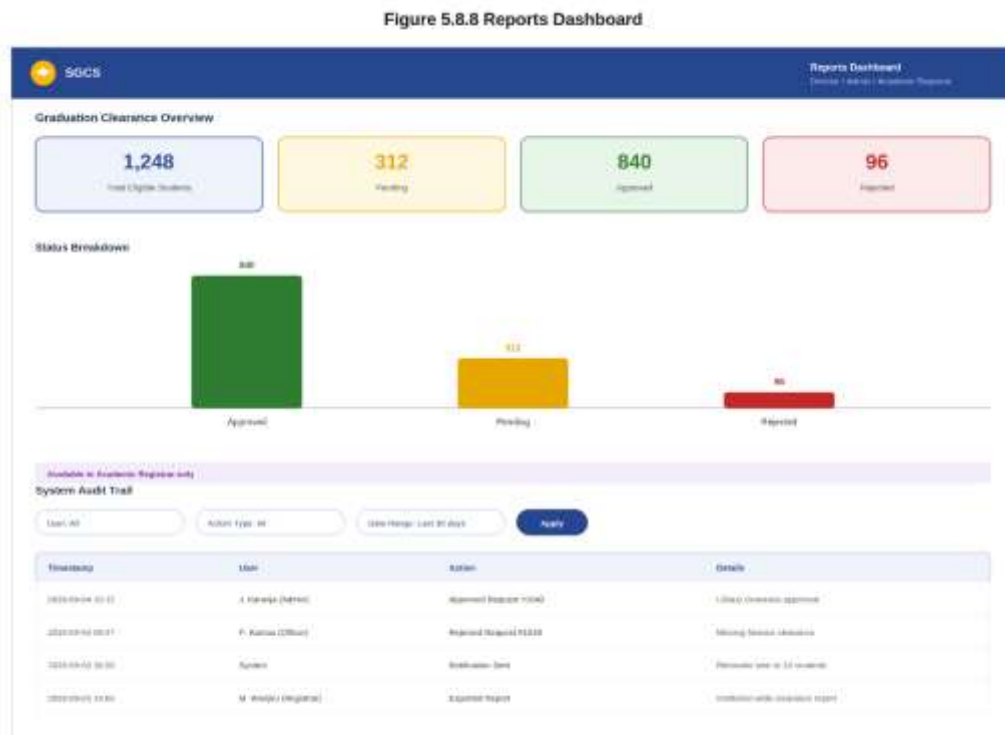
Figure 5.8.4 Officer Dashboard



5.8.8 Reports Dashboard

The reports dashboard provides the Director, Admin, and Academic Registrar with an overview of the graduation clearance process. It displays clearance counts by overall status and supports filtering by status and current department. Director report results are limited to the Director's assigned institute and department. The Academic Registrar and Admin can access the separate audit-trail page.

Figure 5.8.8 Reports Dashboard



5.9 Chapter Summary

This chapter presented the design of the proposed Students' Graduation Clearance System (SGCS), translating the requirements identified in Chapter Four into a blueprint for implementation. It covered the proposed system architecture, process models, database design, and user interface design. The chapter also presented the ERD, database normalization to Third Normal Form (3NF), and the design of key system interfaces, including the landing page, dashboards, clearance tracking, transcript management, approval download, and reports dashboard.

Overall, the system design provides a structured and reliable foundation for implementation by ensuring that all functional requirements are addressed through appropriate architectural, database, and interface designs. The next chapter focuses on the implementation of the SGCS, describing the development and integration of the system components based on the designs presented in this chapter.

CHAPTER SIX: SYSTEM IMPLEMENTATION & TESTING

6.1 Chapter Introduction

This chapter presents the implementation and testing of the proposed Students' Graduation Clearance System (SGCS), demonstrating how the design specifications developed in Chapter Five were translated into a functional web-based application. The system was implemented using HTML, CSS, and JavaScript for the front end, PHP for the back end, and MySQL for database management within the XAMPP development environment. The chapter presents key system interfaces through representative screenshots, describes the functional and non-functional testing conducted to verify that the system meets the specified requirements, and outlines the usability evaluation performed to assess the effectiveness of the system. The chapter concludes with a summary of the implementation and testing outcomes.

6.2 System Screenshots

The screenshots below illustrate the core interfaces of the completed SGCS.

Figure 6.2.1 Landing Page, shows the summary of the system

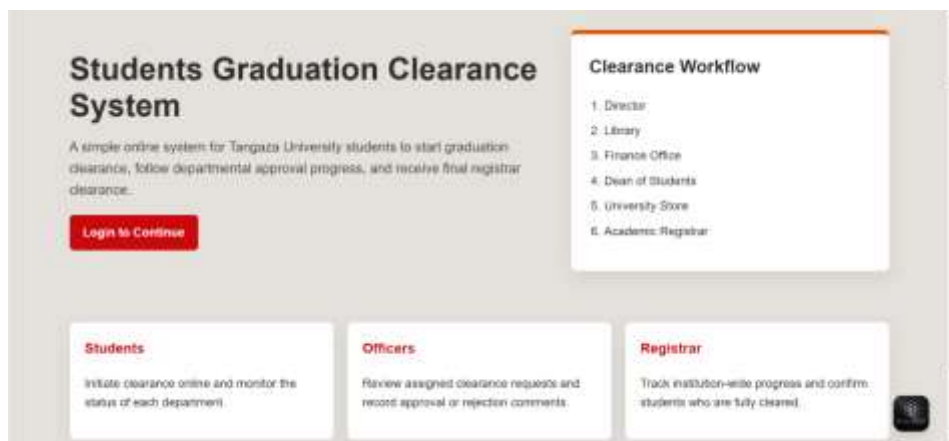


Figure 6.2.2 Login Page, shows the username and password

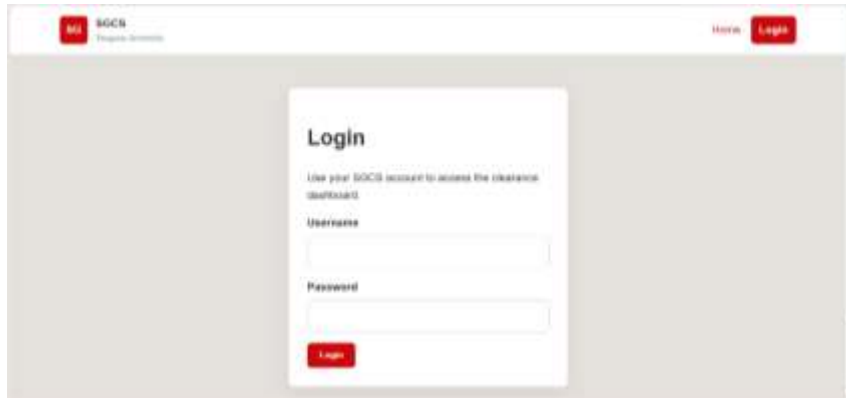


Figure 6.2.3 Student Dashboard page

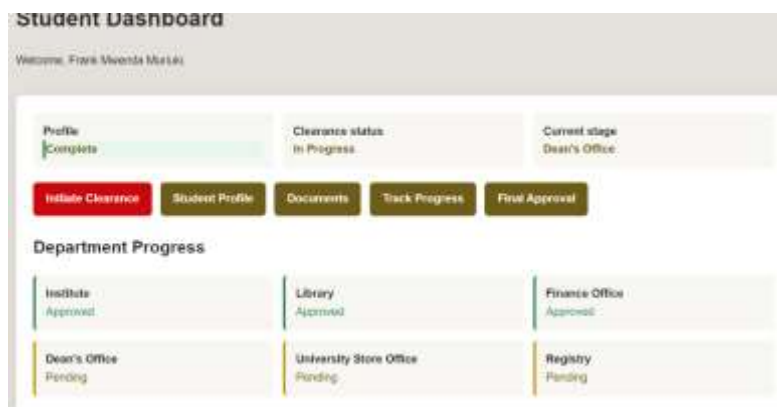


Figure 6.2.4 Officer Dashboard



6.3 Testing Plan

A testing plan is a structured document that outlines the testing activities, objectives, scope, methods, test cases, and expected outcomes used to verify that a system meets its specified requirements.

6.3.1 Unit and Integration testing

Unit testing is a software testing technique that verifies individual system components independently to ensure they function correctly before integration. It was chosen because it

helps identify and correct errors at an early stage of development, improving system reliability and reducing the cost of fixing defects later in the implementation process. For the SGCS, unit testing was conducted on key modules. Each component was tested independently using valid, boundary, and invalid input values before being integrated into the complete system. The testing confirmed that individual modules operated as expected and allowed defects to be identified and corrected before User Acceptance Testing (UAT).

6.3.1 Unit and Integration testing table

Table 6.3.1 Testing Table

Test ID	Module / Requirement	Test Description	Expected Result	Status
TC-01	Login	Attempt login with valid and invalid officer/student credentials.	Valid credentials grant access to the correct role-based dashboard; invalid credentials are rejected with a generic error message.	Pass
TC-02	Clearance Initiation	Student initiates a clearance request from the dashboard.	A new clearance record is created and routed first to the Institute Director for review.	Pass
TC-03	Officer Approval/Rejection	Officer approves a request, and separately rejects one without a comment.	Approval forwards the request to the next department; rejection is blocked until a comment is entered.	Pass
TC-04	Status Tracking	Student views the progress tracker after a departmental decision.	The tracker updates in real time to reflect the current department and status.	Pass
TC-05	Notifications	System generates a notification after a submission, approval, and rejection.	A corresponding notification appears in the student's notification panel for each event.	Pass
TC-06	Financial Verification	Finance Office records an outstanding balance for a student.	The clearance request is held at the Finance stage until the balance is cleared and re-verified.	Pass
TC-07	Reporting Dashboard	Director, Admin and Registrar generate a clearance status report.	The dashboard displays accurate counts of pending, approved and rejected students, filterable by department.	Pass
TC-08	Audit Trail	Officer performs an approval action.	The action, officer, timestamp, and outcome	Pass

Test ID	Module / Requirement	Test Description	Expected Result	Status
			are recorded in the audit log.	

6.3.2 User acceptance testing

User Acceptance Testing (UAT) is the final phase of software testing in which representative end users evaluate the system to determine whether it meets the specified requirements and is suitable for operational use. It was chosen for the Students' Graduation Clearance System (SGCS) because it validates the system from the users' perspective, ensuring that its functionality, usability, and performance meet the needs of students and administrative staff before deployment.

In line with Section 3.7, UAT was conducted with four final-year students and two administrative staff from the Library and Finance Office. Participants performed representative tasks, including logging in, initiating a clearance request, tracking its progress, and approving or rejecting requests. Feedback was collected using a structured evaluation form with ratings from 1 (Very Poor) to 5 (Excellent). Overall, participants found the system more efficient and easier to use than the manual clearance process, highlighting the real-time progress tracker and mandatory rejection comments as valuable features. They also recommended the addition of email notifications alongside the existing in-app notifications as a future enhancement.

Table 6.3.2 User Acceptance Testing

Criterion	Average Score (out of 5)	Comments
Ease of navigating the dashboard	4.2	Participants found the layout intuitive after a brief orientation.
Clarity of clearance status tracker	4.6	Colour-coded stages were consistently well understood.
Usefulness of notifications	4.2	Two officers requested email notifications in addition to in-app alerts.
Ease of approving/rejecting requests	4.5	Mandatory rejection comments were seen as useful, not burdensome.
Overall satisfaction	4.3	All participants preferred the system to the manual process.

6.4 Evaluation Plan

Heuristic evaluation is a usability inspection method in which evaluators assess a system against established usability principles, such as Nielsen's Ten Usability Heuristics, to identify

potential usability issues. It was chosen for the Students' Graduation Clearance System (SGCS) because it provides a systematic and cost-effective approach to evaluating the user interface, allowing usability problems to be identified and resolved before system deployment.

In addition to functional testing, the SGCS was evaluated using Nielsen's heuristic evaluation method by the developer and two participating administrative staff. The evaluation focused on the usability heuristics most relevant to the system and assessed aspects such as interface consistency, user feedback, error prevention, and ease of navigation. The identified usability issues were minor and were resolved before the pilot deployment phase. No critical usability problems that could hinder the effective use or deployment of the system were identified.

Table 6.4: Evaluation Plan

Heuristic	Observation	Action Taken
Visibility of system status	Progress tracker and status badges clearly show the current clearance stage.	No change required.
Error prevention	Rejection could initially be submitted without a comment.	Added mandatory comment validation before submission.
Consistency and standards	Status colours varied slightly between the dashboard and tracking screen.	Standardised colour coding across all screens.
Flexibility and efficiency of use	Frequent officers requested a way to filter pending requests.	Added department-level filtering to the officer dashboard.

6.5 Chapter Summary

This chapter presented the implementation and testing of the Students' Graduation Clearance System (SGCS). It showcased the key system interfaces and described the testing activities, including unit testing, User Acceptance Testing (UAT), and heuristic evaluation. The results confirmed the tested system functions, with only minor defects and usability issues identified and resolved before deployment. Financial clearance approval remains dependent on a future verified fee-data integration. The next chapter presents the project's conclusions, challenges encountered, and recommendations for future improvement.

CHAPTER SEVEN: CONCLUSIONS, FINDINGS AND RECOMMENDATIONS

7.1 Introduction

This chapter concludes the report by reflecting on the extent to which the project achieved its stated aim and specific objectives, summarising the key findings from the design, implementation, and testing of the Students' Graduation Clearance System (SGCS), outlining the challenges encountered during the project, and presenting recommendations for future enhancement of the system.

7.2 Findings and Conclusion

The aim of this project was to design and develop a web-based Students' Graduation Clearance System (SGCS) to automate and streamline the graduation clearance process at Tangaza University. This aim was successfully achieved through the development of a centralized system that replaces the manual, paper-based process.

The study reviewed existing graduation clearance systems, analysed the current manual process, and translated the identified requirements into a functional system design. The SGCS was successfully implemented using HTML, CSS, JavaScript, PHP, and MySQL, incorporating a sequential departmental approval workflow, real-time status tracking, in-app notifications, and reporting features. System testing, including unit testing, User Acceptance Testing (UAT), and heuristic evaluation, confirmed the tested functions. Financial clearance approval requires a future verified fee-data integration. Overall, the SGCS is expected to improve the efficiency, transparency, and management of the graduation clearance process for students and university staff.

7.3 Challenges encountered

1. Limited access to administrative staff: Coordinating interviews, document analysis, and User Acceptance Testing (UAT) sessions with departmental officers was challenging due to their busy schedules.
2. Designing an accurate workflow: Translating the manual clearance process into a structured digital workflow required multiple consultations to ensure it reflected actual institutional procedures.
3. Balancing usability and security: Implementing role-based access control and validation measures while maintaining a user-friendly interface required several refinements based on user feedback.

4. **Limited time for field testing:** The project's academic timeline restricted the duration and scope of pilot testing, limiting the amount of real-world usage data collected.
5. **Single-developer implementation:** Managing all stages of the project, including requirements gathering, design, development, testing, and documentation, within the allocated timeframe required effective planning and time management.

7.4 Future Recommendations

Based on the testing and evaluation results, the following recommendations are proposed for future improvement of the Students' Graduation Clearance System (SGCS):

1. **Full system deployment:** Deploy the system across all six clearance departments and to all final-year students at Tangaza University.
2. **Enhanced notifications:** Integrate email and SMS notifications alongside in-app alerts to keep students informed of important clearance updates.
3. **System integration:** Integrate the SGCS with the university's student records and finance systems to automate academic and financial verification.
4. **Mobile application:** Develop a dedicated mobile application to improve accessibility and convenience for students using smartphones.
5. **Advanced reporting:** Enhance the reporting dashboard with analytics to identify workflow bottlenecks and support informed decision-making.
6. **Expanded usability evaluation:** Conduct a larger-scale usability study involving more students and departmental staff to further evaluate and improve the system after full deployment.

7.5 Summary

This project demonstrated that the graduation clearance challenges at Tangaza University can be effectively addressed through a web-based Students' Graduation Clearance System (SGCS). Developed using standard web technologies and guided by the Structured System Analysis and Design methodology, the system was shown through testing and evaluation to meet its functional and non-functional requirements. Although the implementation was limited by the project's scope and timeframe, it provides a solid foundation for future enhancements and full-scale deployment, supporting a more efficient, transparent, and student-centred graduation clearance process.

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APPENDICES

A: Sample Questionnaire for Final-Year Students

Dear Respondent,

This questionnaire is part of a final-year Computer Science project at the Catholic University of Eastern Africa, which involves developing a Students' Graduation Clearance System for Tangaza University. Completion of this questionnaire should take no more than ten minutes. All information provided will be treated with strict confidentiality and used solely for academic purposes.

SECTION A: BACKGROUND INFORMATION

1. What is your current year of study? ☐ Final Year ☐ Other: _____
2. What is your gender? ☐ Male ☐ Female ☐ Prefer not to say
3. How do you mainly access online services? ☐ Smartphone ☐ Laptop/Desktop ☐ Both

SECTION B: CURRENT CLEARANCE PROCESS

4. Have you gone through the graduation clearance process at Tangaza University? ☐ Yes ☐ No
5. If yes, how long did the entire process take? ☐ Less than 1 day ☐ 1–2 days ☐ 3–5 days ☐ More than 5 days
6. How would you rate the current manual clearance process? ☐ Very Poor ☐ Poor ☐ Fair ☐ Good ☐ Excellent
7. What challenges did you face? (Tick all that apply) ☐ Long queues ☐ Unavailable officers ☐ Multiple return visits ☐ Lost/misplaced forms ☐ Poor communication about status ☐ Other: _____

SECTION C: PROPOSED ONLINE SYSTEM

8. Would you prefer an online clearance system? ☐ Yes ☐ No ☐ Neutral
9. How important are notifications about clearance progress to you? (1 = Not important, 5 = Very important) ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5
10. Which feature is most important to you? (Rank 1–5) ☐ Online initiation ☐ Real-time status per department ☐ Email notifications ☐ Digital certificate download ☐ Automated reminders

11. Do you have reliable internet access? ☐ Yes ☐ Sometimes ☐ No

12. Any other comments or suggestions for the proposed system:

—

Thank you for your participation.

B: Graduation Clearance Form



TANGAZA UNIVERSITY
STUDENT CLEARANCE FORM

STUDENT INFORMATION

Name _____
Student number _____ Institute/Programme _____
Email Address _____ Graduation Year: _____

INSTITUTE CLEARANCE

The above has: ☐ Returned all Material/Does not have any Material that belongs to the University
☐ Material that belongs to the University valued at KShs. _____

Comments: _____

Director's Signature and Stamp: _____

LIBRARY CLEARANCE

The above has: ☐ Returned all Library books/ Material including Library Cards
☐ Library Books/Materials valued at KShs. _____

Comments: - _____

Librarian's Signature and Stamp

OFFICE OF THE DEAN OF STUDENTS

The above has: ☐ Returned all Material/Does not have any Material that belongs to the University
☐ Material that belongs to the University valued at KShs. _____

Comments: - _____

Dean's Signature and Stamp: _____

ADMINISTRATION/UNIVERSITY STORE

The above has: ☐ Returned all Material/Does not have any Material that belongs to the University
☐ Material that belongs to the University valued at Kshs. _____

Comments: -

Administration/Store Signature and Stamp: _____

FINANCE OFFICE

The above has: ☐ No outstanding Debts or Arrears
☐ Outstanding Debts and Arrears Amounting to Kshs. _____

Comments: -

Total Amount due from All Departments/Institutes

Date Cleared: _____

Finance Office Signature and Stamp:

REGISTRAR'S OFFICE

The above has: ☐ Cleared with all the Departments/Institutes and has no Outstanding Balance
☐ Is yet to clear with all the Departments/Institutes

Comments: -

Registrar's Signature and Stamp: _____

Appendix C: Budget Schedule

The estimated budget for this project is presented in Table C1 below. All figures are in Kenyan Shillings (KES). Because all development tools are open-source and freely available,

the primary costs are internet connectivity, web hosting, and documentation. A ten percent contingency allowance has been included to cover unforeseen expenses.

Table C: Project Budget

S/N	Item	Quantity	Unit Cost	Total
1	Laptop Computer (AMD Ryzen 5, 8GB RAM, 256GB SSD)	1	Already owned	KES 0
2	Internet subscription (6 months)	6 months	KES 1,500/month	KES 9,000
3	Web hosting - shared hosting plan (6 months)	6 months	KES 800/month	KES 4,800
4	Domain name registration (1 year)	1	KES 1,200	KES 1,200
5	Printing & binding - proposal + final report	2 copies	KES 1,500	KES 3,000
6	Stationery (notebooks, pens)	1 set	KES 500	KES 500
7	Transport to Tangaza University (interviews + UAT)	4 trips	KES 400/trip	KES 1,600
8	PHP / MySQL / XAMPP / VS Code / Git	Open-source	Free	KES 0
9	Contingency (10%)	—	—	KES 2,010
	TOTAL			KES 22,110

The total estimated budget of KES 22,110 is manageable within the constraints of a student-funded project. The largest single cost is internet subscription, which is essential for research, development, and version control throughout the project period. All development tools are available free of charge, significantly reducing the overall cost of the project.

Appendix D: Time Schedule

The project will run from February 2026 to August 2026, a total of seven months. The Gantt chart below shows when each phase will be active. Some phases overlap intentionally, reflecting the iterative nature of systems development where design and early implementation can proceed concurrently.

Table A1: Time Schedule - Gantt Chart (February to August 2026)

Activity	Feb	Mar	Apr	May	Jun	Jul	Aug
Topic Selection & Proposal Writing	✓	✓					
Literature Review		✓	✓				
Requirement Elicitation			✓				
System Analysis			✓	✓			
System Design				✓	✓		
System Implementation					✓	✓	
System Testing & UAT						✓	✓
Deployment & Report Writing							✓

APPENDIX E: Sample Code

```

* admin_graduation_sets.php
62  if ($_SERVER['REQUEST_METHOD'] === 'POST') {
64      try {
73          if ($action === 'create' || $action === 'update') {
194             if ($action === 'create') {
210                 $stmt->execute([
215                     $send ?: null
216                 ]);
217
218                 $newSetId = (int)$pdo->lastInsertId();
219                 if ($qualificationTypeIds) {
220                     $map = $pdo->prepare('INSERT INTO graduation_set_qualification_types (graduation_set_id, qualification_type_id) VALUES (:set_id, :type_id)');
221                     foreach ($qualificationTypeIds as $qualificationTypeId) {
222                         $map->execute([$newSetId, $qualificationTypeId]);
223                     }
224                 }
225
226                 /* Optional audit log*/
227
228                 if (function_exists('log_audit')) {
229                     log_audit(
230                         $pdo,
231                         $_SESSION['user_id'],
232                         'graduation_set_created',
233                         'Created graduation set: ' . $name
234                     );
235                 }
236
237             }
238         }
239     }
}

```

```

* reports_dashboard.php
5
6  require_login();
7
8  $current = current_user();
9  $role = strtolower(trim((string)$current['role'] ?? ''));
10
11  /* Library, Finance and University Store reports may be used by the dedicated role names as well as
12  $assignedReportDepartment = get_assigned_department($pdo, $current);
13  $assignedReportDepartmentName = strtolower(trim((string)$assignedReportDepartment['dept_name'] ?? ''));
14
15  $isDirector = ($role === 'director');
16  $isSchoolDean = ($role === 'dean');
17  $isGeneralReportUser = in_array($role, ['admin', 'registrar'], true);
18  $isDepartmentReportUser =
19      in_array($role, ['librarian', 'finance_officer', 'university_store'], true)
20      || in_array($assignedReportDepartmentName, ['library', 'finance office', 'university store offi
21
22  if (!$isDirector && !$isSchoolDean && !$isGeneralReportUser && !$isDepartmentReportUser) {
23      http_response_code(403);
24      exit('Not authorized.');
```